EV Pulse NC: Data-Driven EV Infrastructure Gap Analysis for North Carolina

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Date: May 27, 2026

Abstract

North Carolina holds $109 million in federal NEVI Formula Program funding for public EV charging infrastructure but lacks a publicly available, data-driven method for prioritizing county-level allocation against the dual objectives of utilization efficiency and Justice40 equity, against a backdrop of statewide battery electric vehicle (BEV) adoption growing at a 53.8% compound annual rate since 2018. This study integrates five analytical phases into a two-tier NEVI Priority Score formulation combining a county-level ranking with ZIP-level targeting: out-of-sample forecast validation, an updated NREL AFDC infrastructure baseline (1,985 stations, 6,145 connectors), ZIP-level infrastructure-density analysis with an additive Theil-T decomposition, LEHD LODES workplace-charging demand modeling, and a CEJST v2.0 Justice40 equity overlay. Out-of-sample forecast validation across 400 county-month observations produced a mean absolute percentage error of 4.34%, with the 69.00% systematic underprediction explained as the statistical fingerprint of post-IRA adoption acceleration (Chow $F = 1{,}268.35$, $p < 1 \times 10^{-6}$); the Theil-T decomposition across the top-10 county cohort found that 84.5% of charging infrastructure inequality occurs within counties rather than between them, motivating the two-tier architecture. The methodological contribution, to the author's knowledge, is the first additive between/within Theil-T decomposition applied to EV charging infrastructure (extending prior work that used the Theil index as a scalar inequality measure) integrated with a three-pillar scoring framework whose statistical independence (maximum Variance Inflation Factor of 1.41) and triply-robust top-3 stability (Union, Mecklenburg, Guilford) substantiate the rankings under weight perturbation. The framework supplies a prescriptive layer that informs NCDOT's NEVI allocation decisions without substituting for them; the 10-county scoring scope captures 73% of the statewide BEV fleet and is engineered for statewide extension, with all analytical scripts and processed outputs publicly available for verification and reuse.

Keywords: EV charging infrastructure, NEVI Formula Program, Justice40, Theil decomposition, infrastructure equity, North Carolina

1. Introduction

1.1 Overview

This paper presents EV Pulse NC, a data-driven analysis of the alignment between electric vehicle adoption and public charging infrastructure across North Carolina's 100 counties. The forecast validation component achieved a mean absolute percentage error of 4.34% on a four-month out-of-sample holdout across all 100 counties, and the infrastructure data was upgraded from a DCFC-only extract to a full API-based pull covering all charging levels. The ZIP-code infrastructure density analysis identified pronounced within-county inequality using a Theil-T decomposition; the workplace charging demand model used LEHD LODES commuter flow data to quantify daytime demand at employment centers; and a Justice40 equity overlay using the Climate and Economic Justice Screening Tool (CEJST v2.0) integrated equity considerations into the prescriptive output. These five analytical phases (Phases 1 through 5, defined in Section 4.1) converge in a two-tier scoring framework (a NEVI Priority Score at the county level and a ZIP-level equity targeting layer) that ranks the top 10 NC counties by BEV registration count for federal NEVI Formula Program funding consideration.

1.2 Motivation

North Carolina's battery electric vehicle (BEV) fleet grew from 5,165 registered vehicles in September 2018 to 94,371 by June 2025, a 1,727% increase representing a 53.8% compound annual growth rate. The growth has been uneven. Wake County alone accounts for more BEV registrations than the bottom 60 counties combined, and the Gini coefficient for county-level BEV distribution sits at 0.805, a concentration level comparable to highly skewed national wealth distributions.

Public charging infrastructure has not matched this pace. As of January 2026, North Carolina has 1,985 charging stations housing 6,145 individual connectors, producing a statewide ratio of approximately 16.9 BEVs per public charging port, which exceeds the nationally recommended benchmark of 10 to 15. The gap is worse than the statewide average suggests; county-level ratios vary dramatically, and several counties with registered BEVs have zero public charging stations.

The federal government has allocated $109 million to North Carolina through the NEVI Formula Program to expand EV charging infrastructure. How that money gets distributed matters. An allocation driven purely by existing demand would reinforce the geographic concentration already present, funneling investment into the Research Triangle and Charlotte metro while leaving rural eastern North Carolina underserved. An allocation driven purely by equity would place stations where utilization rates may be too low to sustain operations.

This tension between efficiency and equity is not theoretical. In February 2026, NCDOT shifted its NEVI deployment strategy away from a corridor-only approach (50 interstate stations) toward 16 rural and community locations, citing the need to serve underserved areas. The shift validates what this project's Gini analysis quantified: the top 10 counties hold 72% of all BEVs, while the bottom 50 counties account for just 4.8%. The question is not whether to balance efficiency and equity, but how to weigh them when $109 million is on the table and the two objectives point in different directions.

1.3 Goals

First, how accurate are the county-level BEV adoption forecasts produced by SAS Model Studio when tested against data the models have never seen? Forecast validation is a prerequisite for everything else in the project. If the predictions are unreliable, infrastructure planning built on them would be flawed. The out-of-sample validation achieved a mean absolute percentage error of 4.34% across all 100 counties, confirming that the forecasts are usable for planning purposes, though a systematic underprediction bias of 69.00% suggests adoption is accelerating faster than historical patterns indicated.

Second, where are the greatest gaps between EV demand and charging supply at the county level? The baseline relied on a DCFC-only infrastructure extract from July 2024; a full API-based pull from NREL's Alternative Fuels Data Center captured all charging levels and access types as of February 2026, enabling a more complete picture of supply-side coverage.

Third, how do ZIP-code-level infrastructure density patterns within major urban counties reveal deployment priorities that county-level aggregation obscures? Wake County spans 857 square miles. Saying it needs more chargers is not actionable; identifying specific ZIP codes within Wake County that lack coverage is.

The fourth question shifts from where chargers are to who needs them: how do inter-county commuting flows reshape infrastructure demand when workplace charging needs are layered onto residential registration data? The baseline analysis treats BEV demand as a residential phenomenon, which captures roughly 80% of charging behavior but misses employment centers entirely.

Finally, can a weighted scoring equation translate these analytical layers into defensible county-level NEVI funding recommendations? Answering this fifth question motivated the scoring-framework formulation itself, which combines explicit prioritization criteria spanning equity, utilization, and cost-effectiveness. The resulting formulation, `NEVI Score = w1*Equity + w2*Utilization + w3*Cost_Effectiveness`, ties the analytical outputs together into actionable allocation recommendations.

1.4 Project Background

EV Pulse NC is a sole-authored research project conducted at Fayetteville State University's Broadwell College of Business and Economics, completed under the supervision of Dr. Majed Al-Ghandour as faculty advisor. The project is self-directed with no corporate sponsor and operates entirely on publicly available data sources.

The work builds on a baseline analysis completed during Fall 2025 (the *nc-ev-atlas* project). That baseline processed 8,200 county-month observations spanning 100 counties across 82 months of NCDOT registration data on the demand side, paired with 1,725 individual charging connectors across 355 stations from the Department of Energy's Alternative Fuels Data Center on the supply side. The baseline established county-level ARIMA forecasts with a weighted in-sample MAPE of 2.73%, identified the infrastructure gap, and revealed the extreme geographic concentration reflected in the 0.805 Gini coefficient. The baseline results held up well. As the work was extended, however, the gaps became obvious: county-level aggregation masked intra-county variation; the infrastructure snapshot was already 18 months old; the forecasts had never been tested against data they had not seen; and the demand model treated every BEV as a residential vehicle.

This study extends the baseline through five analytical phases plus a scoring integration layer. Phase 1 validated SAS Model Studio's BEV adoption forecasts against four months of held-out data (MAPE 4.34%). Phase 2 upgraded the supply-side infrastructure baseline from a DCFC-only July 2024 extract to a full February 2026 AFDC API pull covering all charging levels and access types. Phase 3 measured ZIP-level infrastructure inequality with both Gini-scalar and additive Theil-T decomposition. Phase 4 added workplace charging demand using LEHD LODES origin-destination commuter flows. Phase 5 layered a Justice40 equity overlay using CEJST v2.0 designations. The NEVI scoring framework integrates the outputs of all five phases into the two-tier prescriptive recommendation layer that anchors this study. Each phase is described in detail in Section 4.1.

One limitation worth flagging upfront: the project operates entirely with public data. ZIP-code-level BEV registration data is not available from NCDOT, commercial alternatives require expensive licenses, and the LEHD LODES 2021 commuting data predates fuller post-COVID return-to-office patterns. Where data quality is strong, the analysis makes specific claims; where it is not, the analysis reports ranges and acknowledges uncertainty. The extensions are designed with this reality in mind, applying strategic selectivity rather than attempting precision the data cannot support.

2. Problem Definition

2.1 Statement of the Problem

North Carolina has $109 million in federal NEVI Formula Program funding to deploy EV charging infrastructure, and no publicly available, data-driven method for deciding which counties should receive it. The state's BEV fleet has grown at a 53.8% compound annual rate since 2018, but that growth is concentrated so heavily that the Gini coefficient across counties reaches 0.805. Charging infrastructure is both insufficient at the statewide level (16.9 BEVs per port versus a 10-to-15 benchmark) and distributed in ways that reflect existing demand rather than projected need. The result is a resource allocation problem where efficiency and equity pull in opposite directions, and where the default outcome, absent analytical intervention, is continued concentration of infrastructure in the Research Triangle and Charlotte metro areas.

The gap is not merely quantitative. NCDOT's February 2026 pivot from a corridor-only NEVI strategy to one that includes 16 rural and community locations signals that policymakers recognize the tension but lack a systematic method for resolving it. This project attempts to supply that method.

2.2 Scope and Boundaries

The analysis covers all 100 North Carolina counties. On the demand side, the temporal range spans September 2018 through October 2025: 86 months of NCDOT BEV registration data comprising 8,600 county-month observations (8,200 in the original baseline plus 400 from the four-month validation holdout). On the supply side, infrastructure data comes from the NREL AFDC API as of February 2026: 6,145 connectors across 1,985 stations covering all charging levels and access types. This replaced the baseline's DCFC-only extract (1,725 connectors across 355 stations from July 2024), which captured fast-charging ports but missed 82.4% of stations that offer only Level 2 charging.

Three analytical scales structure the work. County-level analysis provides the primary unit for NEVI allocation recommendations. ZIP-code-level analysis targets 134 urban ZIP codes within the top 10 counties (representing 73% of the statewide BEV fleet) to identify intra-county deployment priorities that county aggregation masks. Census tract-level data enters through the CEJST equity overlay, which maps Justice40 disadvantaged community boundaries onto the infrastructure data.

The temporal scope of the extensions varies by data source. Forecast validation tests SAS Model Studio predictions against July through October 2025 actuals. The AFDC infrastructure pull reflects station status as of February 2026. LEHD LODES commuting flow data reflects 2021 origin-destination patterns, adjusted for post-COVID remote work shifts using a uniform multiplier. These different temporal windows are a constraint, not something the analysis can smooth over.

2.3 Analytical Complexity

What makes this problem analytically difficult is that it operates across multiple dimensions simultaneously. A county's priority for NEVI funding depends on at least four things: how large its demand-supply gap is, how fast that gap is growing, whether it contains disadvantaged communities that satisfy the Justice40 mandate (40% of benefits directed to underserved populations), and whether workplace commuting flows create demand invisible to residential registration data. No single metric captures all of these.

The scoring equation, `NEVI Score = 0.40*Equity + 0.35*Utilization + 0.25*Cost_Effectiveness`, is a deliberate attempt to compress these dimensions into a single ranked output. The weights are not arbitrary. Equity receives the highest weight (0.40), an analytical anchor reflecting the Justice40 40%-of-benefits target established under Executive Order 14008 (January 2021). The 0.40 weight should be read as informed by, rather than mandated by, current federal policy: EO 14008 was rescinded in January 2025, and while the IIJA-specific NEVI program rule and the federally-approved state NEVI deployment plans retain meaningful-public-involvement and Justice40-aligned considerations, the explicit 40%-benefit-flow target no longer carries Executive Order standing. Utilization receives 0.35 because validated BEV forecasts represent the strongest data in the project. Cost-effectiveness receives the lowest weight (0.25) because workplace charging estimates carry the most uncertainty, relying on 2021 LODES commuting data that predates fuller post-COVID return-to-office and may overstate flows by 10 to 25%.

The component scores travel with the composite, so a county might rank 15th overall but 3rd on equity and 60th on utilization; nothing is hidden. Still, any weighting scheme embeds assumptions about what matters most, and reasonable people could disagree on whether equity deserves 0.40 or 0.50.

2.4 Success Criteria

The hardest bar to clear is practical defensibility. The top-ranked counties should make intuitive sense to someone familiar with North Carolina's EV environment, and when they do not, the scoring components should explain why. That matters more than methodological rigor on its own, though the methodology still has to hold up: validated forecasts, current infrastructure data, and federal equity mandates all have to underpin the output. The deliverable itself is a ranked list of the top 10 NC counties with composite NEVI priority scores, component-level breakdowns, and a framework designed for statewide extensibility to the remaining 90 counties.

An optional external check via NCDOT's actual February 2026 deployment decisions provides validation: agreement between this study's equity-weighted rankings and NCDOT's planned 16 rural/community locations would suggest the methodology captures real policy priorities. Disagreement would be equally informative, revealing where the scoring weights diverge from revealed state preferences.

2.5 Exclusions

The project does not address site-level placement. Identifying that ZIP code 27603 in Raleigh needs charging infrastructure is within scope; selecting the specific parking lot or retail location is not. Private and residential charging, which accounts for roughly 80% of all EV charging, falls outside the analysis entirely because the NEVI program funds public infrastructure. Real-time utilization data (how often existing chargers are actually used) is not publicly available at the station level, so the analysis relies on BEV-to-port ratios as a proxy for demand pressure rather than measured throughput. The geographic boundary is North Carolina; cross-state commuting effects from Virginia and South Carolina are not modeled, though they likely affect border counties. Finally, the project does not model the political dynamics of NEVI fund distribution. The scoring equation optimizes for a combination of equity, utilization, and cost-effectiveness; actual allocation decisions involve legislative priorities and stakeholder negotiations that this project cannot model.

3. Literature Review

3.1 EV Adoption Trends and Geographic Concentration

The growth trajectory of electric vehicle adoption in the United States has been well documented, but the spatial distribution of that growth has received less attention than it deserves. National BEV registrations surpassed 1.4 million in 2023, up from fewer than 100,000 in 2015, yet the growth is geographically uneven in ways that complicate infrastructure planning (DOE, 2024). California alone accounts for roughly 40% of national BEV registrations, and within states the concentration pattern repeats at finer scales. North Carolina illustrates this clearly: the state's BEV fleet grew at a 53.8% compound annual rate between 2018 and 2025, but Wake County holds more registrations than the bottom 60 counties combined, producing a county-level Gini coefficient of 0.805 (NCDOT, 2025). That level of concentration is comparable to highly skewed national wealth distributions, and it creates a fundamental tension for infrastructure planners: invest where the vehicles already are, or invest where equitable access demands they should be.

The Joint Office of Energy and Transportation reported that public charging infrastructure nationally grew from approximately 50,000 stations in 2020 to over 186,000 by late 2024, but that the ratio of BEVs to public charging ports has worsened as vehicle sales outpace charger deployment (Joint Office, 2024). North Carolina's ratio of 16.9 BEVs per public port exceeds the DOE's recommended 10-to-15 benchmark, and the statewide figure obscures county-level variation that ranges from single-digit ratios in university towns like Orange County to over 100:1 in suburban growth corridors like Union County. This pattern of aggregate sufficiency masking local inadequacy recurs throughout the literature and motivates the sub-county analytical approach this project takes.

3.2 Infrastructure Gap Measurement

Quantifying the mismatch between EV demand and charging supply has been approached through several methodological traditions. The facility location literature, rooted in the p-median and maximal coverage location problems first formalized by Hakimi (1964) and ReVelle and Swain (1970), treats charger placement as an optimization problem: given a set of demand points and candidate locations, minimize travel distance or maximize population coverage. Celik and Ok (2024) extended this tradition by coupling a genetic algorithm solution to the p-median problem with discrete-event simulation to jointly optimize station locations and charging unit capacities. Their work highlights a gap that persists across the optimization literature: models typically assume fixed demand and do not account for the temporal evolution of EV adoption or the heterogeneity of charging needs across vehicle populations.

NC-specific research has addressed pieces of this problem. A Raleigh-focused study demonstrated that solving a maximal coverage location problem using existing gas station sites as candidates improved fast-charger demand coverage by over 50% compared to the existing charger network (Zafar, Bayram, & Bayhan, 2021). The GreenEVT testbed at UNC Greensboro coupled traffic microsimulation with distribution grid simulation to evaluate charging impacts at the household level, an unusual level of spatial resolution (Nilsson, Owen Aquino, Coogan, & Molzahn, 2024). A Cary utilization study analyzed three years of session logs from six public chargers and found that Sunday charging dominated usage patterns, with per-minute energy consumption between 87 and 111.3 Wh, rising steadily from 2019 to 2022 (Kutela et al., 2024). These studies contribute spatial optimization, grid-aware siting, and temporal demand characterization, respectively, but none addresses the distributional equity question: not just whether chargers are optimally placed, but whether the resulting distribution is fair across population groups.

This project fills that gap by treating infrastructure distribution as an inequality measurement problem rather than a pure optimization problem. The BEV-to-port ratio serves as the primary demand-supply metric, calculated at both county and ZIP-code levels. But ratios alone do not capture how evenly or unevenly charging access is distributed within a geographic unit, which is where inequality indices become necessary.

3.3 Inequality Measurement: Gini and Theil Decomposition

The Gini coefficient is the standard tool for measuring distributional inequality and has been applied to income, wealth, health access, and environmental burden distributions for decades. Its application to infrastructure access is more recent. Studies have used Gini coefficients to measure inequality in broadband access, public transit coverage, and water infrastructure distribution, establishing a precedent for treating physical infrastructure allocation as a distributional fairness question rather than a pure efficiency question.

This project applies the Gini coefficient to EV charging port density across ZIP codes within each county, producing values that range from 0.408 (Durham, the most equal among the top 10 counties) to 0.623 (Mecklenburg, the most unequal). The statewide population-weighted Gini of 0.566 confirms that charging infrastructure in North Carolina is distributed with very high inequality, though still less concentrated than BEV ownership itself (Gini 0.805). The Gini captures the overall shape of the distribution, but it cannot answer a question that matters for policy: how much of the inequality occurs between counties versus within them?

The Theil index, specifically the Theil-T (GE(1)) measure developed by Theil (1967) and formalized for subgroup decomposition by Shorrocks (1980), can answer that question. Unlike the Gini, the Theil index is exactly decomposable into between-group and within-group components, meaning total inequality can be partitioned without residual. This property has made it the preferred tool for decomposing income inequality across regions, racial groups, and sectors in economics. It has been applied to health care access, educational attainment, and energy consumption inequality.

Prior EV charging equity studies have used the Theil index as a scalar inequality measure alongside Lorenz curves, Gini coefficients, and the Palma ratio. Choi, Xu, and Jiao (2025), publishing in *Transportation Research Part D*, applied this toolkit to EV charger accessibility in Austin, Texas, reporting Theil as a single magnitude alongside the other metrics. What has not been done, as far as the author's literature search can determine, is exploiting the **additive decomposability of the GE(1) class** (Bourguignon, 1979; Shorrocks, 1980) to partition EV charging infrastructure inequality into its between-county and within-county components. This is a methodological distinction worth naming: a scalar Theil reports the *magnitude* of inequality, while a Theil-T decomposition reports its *structure*: between which units versus within which units the inequality lives. The structural question matters for allocation policy in a way the scalar version does not. This project's Theil-T decomposition produced a striking result: 84.5% of charging infrastructure inequality in the top 10 NC counties is within-county, not between-county. A robustness check using Theil-L (GE(0)) confirmed the pattern at 82.5% within-county. The exact decomposition verified to machine precision, with a difference of 2.22e-16 between the sum of components and total inequality.

The policy implication is direct. County-level allocation formulas, which is how most state NEVI programs currently distribute funding, would miss the majority of the inequality problem. A county like Mecklenburg appears adequately served in aggregate, but Charlotte

Uptown (ZIP 28202) has 78.64 ports per 10,000 people while Charlotte East (ZIP 28215) has 0.31, a 250-fold gap within the same county. The Theil decomposition provides the statistical justification for a two-tier scoring approach: county-level rankings identify where to invest, while ZIP-level targeting identifies for whom.

3.4 Justice40 and the CEJST Policy Context

The federal Justice40 Initiative, established by Executive Order 14008 in January 2021, set a target that 40% of the benefits of certain federal investments flow to disadvantaged communities. The NEVI Formula Program, established under the Infrastructure Investment and Jobs Act of 2021 (IIJA Section 11401; codified at 23 U.S.C. §175), fell under this Justice40 target during the period in which EO 14008 was in force. Executive Order 14008 was rescinded in January 2025, removing the explicit 40%-benefit-flow target from federal executive policy. The IIJA-specific NEVI program rule and the FHWA-approved state NEVI deployment plans retain Justice40-aligned considerations (notably the meaningful-public-involvement requirement and the equity expectations embedded in the NEVI deployment plan template), but the explicit 40% target no longer carries Executive Order standing. This study treats the 0.40 equity weight in its scoring framework as an analytical anchor informed by the prior federal benchmark, not as a current federal mandate.

The operational definition of "disadvantaged" comes from the Climate and Economic Justice Screening Tool (CEJST v2.0; Council on Environmental Quality, 2024), which flags Census tracts that are both low-income (at or above the 65th percentile) and overburdened on at least one of eight environmental, health, or infrastructure dimensions. This is not a synonym for poverty; a tract must experience a measurable burden in addition to economic hardship to qualify.

CEJST uses a "one-strike" design that intentionally casts a wide net: exceeding the 90th percentile on any single indicator across eight burden dimensions spanning environmental, health, and infrastructure indicators triggers the designation, provided the low-income threshold is also met. Seven of the eight categories rely on empirical federal datasets (CDC health surveillance, Census ACS, EPA facility databases, DOE utility cost records). The climate change category is the only one incorporating forward-looking model projections alongside historical data from the FEMA National Risk Index.

In North Carolina, 934 of 2,170 evaluable Census tracts (43.0%) are designated disadvantaged, substantially above the national average of approximately 37%. The elevated rate reflects the state's mix of rural poverty in eastern NC and Appalachia, above-average health burdens, housing cost stress, and legacy industrial contamination. For the project's top 10 BEV counties, the population-weighted disadvantaged rate drops to 18.5%, reflecting that these are North Carolina's wealthiest urban and suburban counties. But nearly one in five residents still lives in a disadvantaged tract, which reinforces the within-county inequality finding from the Theil decomposition.

One complication deserves mention. CEJST was removed from `screeningtool.geoplatform.gov` on January 22, 2025 following the rescission of Executive Order 14008. Data scientists from the EDGI / Public Environmental Data Partners coalition archived and restored the data within days, and this project uses the community-preserved CEJST v2.0 (December 2024 release). The CEJST methodology itself is independently archivable through the CEQ Technical Support Document, which remains the federal record of the screening methodology. The data integrity has been verified against pre-removal records, and the methodology document anchors the analytical operation; the archival pathway is a data-continuity matter rather than a methodological one. No federal successor instrument exists at the time of this writing. EPA EJScreen, the most natural candidate, uses different categories and thresholds and is not a drop-in replacement.

Federal infrastructure programs increasingly rely on multi-criteria scoring frameworks to prioritize discretionary investments. The USDOT Rebuilding American Infrastructure with Sustainability and Equity (RAISE) program scores applications across project outcomes, project readiness, equity, environmental sustainability, and quality of life, with Justice40 considerations embedded in the equity sub-criterion. The BIL Charging and Fueling Infrastructure Discretionary Grant (CFI) program similarly applies multi-criteria evaluation to community charging investments, weighting deployment readiness, equity, and corridor strategy. This study's NEVI Priority Score formulation (equity, utilization, and cost-effectiveness pillars) sits within this established federal practice of multi-criteria infrastructure scoring rather than imposing a novel framework. The methodological contribution claimed here is not the multi-criteria structure itself, which has federal precedent, but the integration of an additive Theil-T decomposition diagnostic with the scoring architecture (Section 3.3) and the statewide-extensible two-tier design.

3.5 Workplace Charging and Commuting Demand

Most EV infrastructure research treats charging demand as a residential phenomenon: vehicles are registered at home addresses, so chargers should be placed near where people live. This captures roughly 80% of charging behavior but misses a critical dimension. Employment centers generate charging demand from commuters who drive in from surrounding counties, and that demand is invisible to residential registration data.

The Census Bureau's LEHD LODES dataset provides origin-destination commuter flows at the Census block level, enabling measurement of how many workers cross county boundaries daily. For North Carolina, Mecklenburg County absorbs a net +194,361 commuters per day, Wake County +126,517, and Durham +89,450 (LEHD LODES, 2021). These are not residents; they are workers who arrive, park for 8 to 10 hours, and leave. The parking duration makes workplace charging a natural fit for Level 2 infrastructure, where 4 to 8 hours at 7 kW can deliver a full charge without the capital cost of DC fast chargers.

The literature on workplace charging is thinner than might be expected. Uimonen and Lehtonen (2020) simulated charging station load profiles in office buildings based on occupancy data and found that workplace charging demand peaks mid-morning and remains elevated through the afternoon, a temporal pattern that complements rather than competes with residential evening peaks. The UCLA Luskin Center's DCF model for EV charging stations (Chang, Erstad, Lin, Rice, & Tsao, 2012) included workplace scenarios and found that workplace installations achieve positive NPV faster than standalone public chargers due to predictable demand from repeat users. But few studies have attempted to estimate workplace charging demand at scale using commuter flow data, and none that this review identified has done so for North Carolina.

This project uses LEHD LODES 2021 origin-destination flows filtered through a three-layer adjustment pipeline: SE03 income segmentation (workers earning above $40,000), an ACS-derived correction factor to estimate the proportion of households above $75,000 (the approximate EV affordability threshold), and a 0.85 remote work reduction multiplier based on Barrero, Bloom, and Davis (2023). The result is an adjusted workplace charging demand estimate of 859,260 statewide, which feeds into county-level port requirement calculations and ultimately into the cost-effectiveness pillar of the NEVI scoring equation.

A limitation of this approach is that LODES reports 2021 commuting patterns, which may not fully reflect post-COVID hybrid work arrangements. The 15% remote work reduction is a national average; sector-specific and geography-specific rates could differ. The project reports results at baseline (30% charging adoption rate) and under low/high scenarios to bracket this uncertainty.

3.6 Scoring and Prioritization Precedents

Multi-criteria infrastructure prioritization is well established in transportation planning. State departments of transportation routinely score bridge and road projects using weighted composites of condition, traffic volume, safety risk, and economic impact. The approach has been applied to EV infrastructure more recently. NREL projected national infrastructure investment needs by region using demand modeling and cost estimation (Wood, Rames, Muratori, Srinivasa Raghavan, & Melaina, 2017), though the analysis did not produce county-level prioritization scores. The DOE's Alternative Fuels Data Center provides station-level economic tools, but these operate at the individual station level rather than the geographic allocation level.

What is less common is combining equity metrics with utilization and cost-effectiveness into a single scoring equation for geographic prioritization. The NEVI program's Justice40-aligned framing creates the need for such a composite, but few published studies have operationalized it. This project's scoring equation, NEVI Score = 0.40 × Equity + 0.35 × Utilization + 0.25 × Cost-Effectiveness, takes that step explicitly. The weight rationale is discussed in Section 2.3; the key point for the literature context is that the weights reflect data confidence levels, with equity highest due to the Justice40-aligned federal benchmark and cost-effectiveness lowest due to the LODES 2021 commuting data uncertainty.

Weight sensitivity analysis tested equity weights from 0.30 to 0.50 in 0.05 increments while holding the utilization-to-cost-effectiveness ratio constant; this is the first of three robustness checks developed in Section 4.7.1 (the "triply-robust" framing). The top three counties (Union, Mecklenburg, Guilford) remained stable across all five scenarios, though their internal ordering shifted: at higher equity weights, Mecklenburg rises and Union drops, reflecting their different pathways to high scores. This stability suggests the rankings are not artifacts of a particular weighting scheme, though the analysis covers only the top 10 counties by BEV count and would need to be extended to all 100 for statewide applicability.

3.7 Synthesis and Gap

The literature provides mature tools for EV charger siting (facility location models), established methods for measuring distributional inequality as a scalar magnitude (Gini coefficients, Theil index as in Choi et al., 2025), a federal policy framework for equity (Justice40 / CEJST, with the post-rescission caveats discussed in Section 3.4), and growing recognition that workplace commuting creates charging demand invisible to residential registration data. What is missing is a study that integrates all of these into a single analytical pipeline for a specific state's NEVI allocation decisions. The methodological contribution this study claims is twofold. First, it applies the additive decomposability of the GE(1) class (Bourguignon, 1979; Shorrocks, 1980) to EV charging infrastructure inequality. This is a *structural* decomposition; prior work using Theil as a *scalar* (Choi, Xu, & Jiao, 2025) does not perform it. Second, it integrates that decomposition into a two-tier allocation framework that combines a county-level NEVI Priority Score with ZIP-level equity targeting. The combination of residential demand, workplace commuting flows, federal equity designations, and validated forecasts into a weighted scoring equation has not been published for North Carolina or, as far as this review can determine, for any other state. This study does not claim to solve the NEVI allocation problem; it claims to provide a data-driven starting point that no one has published for North Carolina yet.

4. Methodology

4.1 Analytical Pipeline

The study executes five analytical phases that integrate into a single prescriptive scoring framework. Each phase is scoped to a specific analytical commitment with defined inputs, outputs, and validation evidence:

- **Phase 1 — Forecast validation.** Out-of-sample testing of SAS Model Studio BEV adoption forecasts against July through October 2025 actuals (Section 6.1).
- **Phase 2 — Infrastructure baseline.** Upgrade from a DCFC-only July 2024 extract (355 stations, 1,725 connectors) to a full February 2026 AFDC API pull (1,985 stations, 6,145 connectors) (Section 6.2).
- **Phase 3 — ZIP-level inequality.** Within-county infrastructure-density analysis using Gini coefficient and additive Theil-T decomposition (Sections 4.4, 6.3).
- **Phase 4 — Workplace charging demand.** Origin-destination commuter flow modeling using LEHD LODES 2021 (Section 4.6).
- **Phase 5 — Justice40 equity overlay.** Area-weighted tract-to-ZCTA crosswalk integrating CEJST v2.0 disadvantaged-community designations (Section 4.5).

The NEVI scoring framework integrates the outputs of all five phases into a composite Priority Score with three pillars (Equity, Utilization, Cost-Effectiveness):

NEVI Score = 0.40 × Equity + 0.35 × Utilization + 0.25 × Cost-Effectiveness

The weight rationale is discussed in Section 2.3. Sensitivity analysis tested equity weights from 0.30 to 0.50 in 0.05 increments; the top three counties (Union, Mecklenburg, Guilford) remained stable across all scenarios, though their internal ordering shifted in expected ways.

The five phases are not strictly sequential. Exploratory profiling of the LEHD LODES commuter data in Phase 4 revealed that the income segmentation threshold needed correction before any downstream analysis could proceed, which sent the pipeline back through a data enrichment step using ACS tract-level income distributions. The Phase 1 validation findings (systematic underprediction, 69.00% of forecasts below actuals) propagated forward to inform how forecast-based infrastructure allocations are buffered in the scoring framework. The five-phase label describes the overall arc, not a rigid sequence.

4.2 Demand-Side Analysis

The demand side of the analysis rests on NCDOT's complete BEV registration time series. The baseline dataset comprises 8,200 county-month observations spanning 100 counties across 82 months (September 2018 through June 2025). The validation extension added 400 observations from the July-through-October 2025 holdout period, bringing the total to 8,600 county-month records.

County-level ARIMA forecasting was performed in SAS Model Studio, which auto-selected the best model type per county: exponential smoothing for 82 counties, ARIMA for 13, and unobserved component models for 5. The forecasts project BEV adoption through June 2028 with a weighted in-sample MAPE of 2.73%. Out-of-sample validation against the four-month holdout produced a MAPE of **4.34%**, below the 5% threshold that the project treats as the usability cutoff.

The validation also uncovered a systematic bias. Actual registrations exceeded forecasts in **69.00%** of observations, with a mean bias of **+18.22 vehicles per county-month**. The 95% prediction interval achieved only **75.50%** raw coverage, well below nominal; after county-specific bias correction, coverage recovers to **93.75%** (a decomposition discussed in detail in

§4.9 below). The underprediction was most pronounced in high-growth urban counties; Mecklenburg County's October 2025 actual exceeded its forecast by 975 vehicles. This pattern is consistent with an adoption curve that has steepened since the Inflation Reduction Act tax credits took effect, and it means forecast-based infrastructure planning needs an upward buffer of 4 to 5% to avoid systematic underinvestment.

One methodological detail that only became apparent during validation: NCDOT changed its registration counting methodology after May 2025. A binary flag (`Methodology_PostMay2025`) was added to the dataset to mark the break. The change did not require separate exploratory analysis because the underlying data structure remained identical to the baseline, so the existing pipeline handled it without modification. Empirical analysis places the change's magnitude at approximately 0.4% of statewide totals (370 vehicles), contributing roughly 20% of the observed underprediction bias. The direction of the change *lowers* counts: the new methodology deduplicates records and excludes recently removed plates. This means the change works *against* the underprediction direction, making the validation a stricter test than the prior methodology would have produced.

4.3 Supply-Side Analysis

The baseline analysis from Fall 2025 used a manually exported AFDC dataset filtered to DC fast chargers only: 1,725 connectors across 355 stations. A full programmatic pull from the NREL AFDC API replaced that extract, capturing 6,145 connectors across 1,985 stations in all charging levels (Level 1, Level 2, and DCFC) and all access types (1,876 public, 109 private). The API-based approach solved two problems at once: it eliminated the six-month data staleness that had accumulated since the July 2024 extract, and it broadened the scope from DCFC-only to the complete infrastructure picture. All subsequent analysis uses the February 2026 API pull exclusively.

The difference between the two datasets illustrates why the upgrade mattered. The DCFC-only extract captured nearly all fast-charging ports (1,742 of 1,745 statewide) but missed 4,298 Level 2 ports and all 35 Level 1 ports. Since Level 2 chargers account for 82.4% of all stations and are the natural fit for workplace charging (4 to 8 hours at 7 kW matches a typical workday), any infrastructure gap analysis built on DCFC-only data would systematically misrepresent supply in exactly the counties where workplace charging demand is highest.

Supply-side analysis operates at the station level in the February 2026 dataset, with port counts stored in aggregate columns per charging level. A station with two 7 kW Level 2 chargers delivers fundamentally different capacity than a station with eight 250 kW DC fast chargers, yet most published EV infrastructure studies count stations rather than connectors. The connector-level granularity matters especially for capacity-weighted BEV-to-port ratios and for cost-effectiveness estimates that depend on throughput potential.

4.4 Inequality Measurement

Infrastructure inequality is measured through two complementary indices applied at the ZIP code level within each county. The Gini coefficient captures the overall shape of the distribution: a county-level Gini of 0.623 for Mecklenburg indicates that charging access varies enormously depending on which ZIP code a resident lives in. The statewide population-weighted Gini of 0.566 sits below the 0.805 Gini for BEV ownership concentration, suggesting infrastructure is partially tracking demand but not doing so equitably.

The Gini, however, cannot decompose inequality into its spatial components, which is where the Theil index enters. The Theil-T (GE(1)) measure, following the decomposition methodology formalized by Shorrocks (1980) and the broader additive-decomposability framework of Bourguignon (1979), partitions total inequality into between-county and within-county components with no residual. The decomposition verified to machine precision (difference of 2.22e-16 between summed components and the total). A robustness check using Theil-L (GE(0)) confirmed the pattern.

The result was unambiguous: 84.5% of charging infrastructure inequality in the top 10 NC counties occurs within counties, not between them. The policy implication is discussed in Section 3.3 and directly shapes the scoring architecture. County-level allocation formulas, which is how most state NEVI programs currently distribute funding, would miss the majority of the inequality problem.

It is worth being explicit about what role the Theil decomposition plays in the scoring framework. **The Theil decomposition is a diagnostic input to architectural choice, not a parameter input to either tier's score.** Neither the NEVI Priority Score (Tier 1, the county-level ranking) nor the ZIP-level equity score (Tier 2, the sub-county targeting layer) takes Theil components as variables. Tier 1 uses equity, utilization, and cost-effectiveness pillars; Tier 2 uses population, disadvantaged-community status, and station density. The 84.5% within-county finding justifies the existence of Tier 2 (without it, a county-only allocation would appear sufficient), but does not enter either tier's computation as an input.

4.5 Equity Overlay

This study's CEJST integration follows the methodological standard established by EPA EJScreen and the HUD USPS Crosswalk for area-weighted spatial integration of federal equity data. (The HUD USPS Crosswalk is the federal-program standard for ZIP-to-Census geographic translation; alignment with this standard pre-empts methodology-substitution challenges from reviewers expecting a different crosswalk approach.) The equity dimension integrates Justice40 disadvantaged community designations from CEJST v2.0 (December 2024 release) into the scoring pipeline. As described in Section 3.4, CEJST flags Census tracts that are both low-income and overburdened on at least one of eight environmental, health, or infrastructure dimensions. In North Carolina, 934 of 2,170 evaluable tracts (43.0%) carry the designation, substantially above the national average.

Operationalizing CEJST data required a spatial crosswalk between 2010-vintage Census tracts (the boundaries CEJST uses) and 2020-vintage ZCTAs (the sub-county unit for infrastructure analysis). The crosswalk uses area-weighted interpolation in the NC State Plane projection (EPSG:32119), the same approach used by EPA EJScreen and HUD's USPS Crosswalk. Each ZCTA's disadvantaged population share is estimated as the area-weighted average of the binary disadvantaged flags from its constituent tract fragments. Intersection polygons smaller than 100 square meters were discarded as geometric artifacts, and border-state tracts from Virginia, South Carolina, Tennessee, and Georgia were included to ensure complete coverage of cross-border ZCTAs. Twenty-five zero-population tracts (12 water-body, 13 institutional and military) were excluded from the denominator.

The crosswalk introduces a known limitation: 2010 tract boundaries do not perfectly align with 2020 ZCTA boundaries, and area-weighted interpolation assumes uniform population distribution within tracts. At the county level this is irrelevant because North Carolina's county boundaries are geographically identical between the two Census vintages. At the sub-county level, the error is modest in urban areas (where tracts are small) but larger in rural tracts that span heterogeneous land uses. A dasymetric refinement using population grids would improve precision and is noted as future work.

A sensitivity analysis tested the robustness of the equity scores to CEJST's most debatable component. Removing the climate change category entirely (the only category that incorporates forward-looking model projections) reduced the statewide disadvantaged rate from 43.0% to 37.3%. Seven of ten study counties showed zero impact; the three affected counties (New Hanover, Durham, Wake) shifted by 2.2 to 4.7 percentage points. The equity rankings are not sensitive to this choice.

4.6 Workplace Charging Demand

The baseline analysis treated BEV demand as purely residential, which captures approximately 80% of charging behavior but ignores employment centers entirely. Phase 4 adds workplace charging demand using the Census Bureau's LEHD LODES 2021 origin-destination commuter flows at the Census block level, replacing the originally planned CTPP dataset on three criteria: more recent vintage (2021 versus 2012–2016), finer geography (block-level versus county-level), and free direct download.

The raw LODES data contains 3,768,428 intra-NC commuter flow records. Transforming these into workplace charging demand estimates requires a three-layer adjustment pipeline. The first layer filters for SE03 workers (those earning above $40,000 annually), reducing the pool to 2,041,731. The second layer applies a county-specific ACS income correction factor, calculated as the ratio of households above $75,000 to those above $40,000 in each origin tract, to approximate EV-affordable income levels. This step accounts for the mismatch between LODES individual earnings and ACS household income. The filter reduces the count to 1,010,894. The third layer applies a 0.85 remote work multiplier from Barrero, Bloom, and Davis (2023), producing a final adjusted demand estimate of 859,260 statewide.

An earlier version of the pipeline included a fourth layer adjusting for renter tenure, on the theory that commuters from renter-heavy tracts would have greater workplace charging need due to limited home charging access. This adjustment was removed during advisor review because no NC-specific calibration data exists, and because the national literature on renter EV adoption rates (3 to 4 times lower than homeowner rates) would pull the multiplier in the opposite direction from what the pipeline intended. Renter share data is retained for descriptive reporting and feeds into the equity pillar through CEJST's housing burden category, but it does not enter the demand calculation directly.

Getting the income correction right was harder than expected. LODES reports individual earnings in three coarse bins; ACS B19001 reports household income across 16 bins. The unit mismatch is real: a worker earning $35,000 might live in a $120,000 household. The county-specific correction factor bridges this gap, but it assumes the income composition of commuters matches the income composition of their origin tract's households, which is an approximation. The project reports results at $60,000, $75,000, and $100,000 income thresholds to bracket this uncertainty.

Net commuter flow analysis classified all 100 NC counties into three typologies: 8 employment centers (net inflow above +10,000), 75 balanced counties, and 17 bedroom communities (net outflow below -10,000). Mecklenburg absorbs the largest net inflow (+194,361 commuters per day), followed by Wake (+126,517) and Durham (+89,450). The workplace port estimates that emerge from these flows feed into the cost-effectiveness pillar of the NEVI scoring equation at a baseline 30% charging adoption rate and 15:1 port ratio, with low and high scenarios bracketing the assumptions.

4.7 NEVI Scoring Construction

The scoring equation combines three pillars into a single composite ranking for each county. Each pillar is itself a weighted composite of sub-metrics, all min-max normalized to a 0-to-1 scale.

The equity pillar (weight 0.40) draws on four sub-metrics: Justice40 population-weighted disadvantaged percentage (weight 0.40 within the pillar), population-weighted Gini coefficient (0.30), count of underserved ZIP codes (0.20), and percentage of ZIP codes with zero stations (0.10). The utilization pillar (weight 0.35) draws on two sub-metrics: BEV-to-port ratio from the February 2026 AFDC infrastructure data (primary differentiator) and a forecast underprediction buffer of 0.045 applied globally. The cost-effectiveness pillar (weight 0.25) draws on three sub-metrics: workplace charging efficiency, net commuter demand magnitude, and population density.

Of the nine designed sub-metrics, seven actively differentiate counties in the 10-county study cohort. Two carry zero variance: `zero_station_pct` is 0.0 for all 10 counties because every top-10 urban county has at least some stations in every inhabited ZIP, and `forecast_buffer` is a uniform 0.045 applied globally rather than per-county. Both would contribute signal if the scoring were extended to all 100 counties, where rural counties have zero-station ZIPs and county-specific forecast errors could replace the global buffer. The effective scoring resolution for the 10-county analysis relies on three equity sub-metrics (`justice40_pct`, `gini_weighted`, `underserved_zips`), one utilization sub-metric (`bev_per_port`), and three cost-effectiveness sub-metrics (`workplace_efficiency`, `commuter_demand`, `pop_density`).

Union County's utilization score illustrates a compression problem inherent in min-max normalization. At 101.5 BEVs per port, Union is more than three times the next-highest county

(Cabarrus at 32.0). Normalizing to the 0-to-1 scale makes Union 1.0 and compresses the remaining nine counties into the lower portion of the utilization scale, reducing the utilization pillar's ability to differentiate mid-ranked counties. The deficit is genuine, not a data artifact, but it is worth acknowledging that the scoring is sensitive to extreme values in small-n cohorts.

The final rankings place Union first (0.561), Mecklenburg second (0.548), and Guilford third (0.465). These three represent distinct archetypes: Union is utilization-driven (suburban growth outpacing infrastructure), Mecklenburg is equity-driven (concentrated disadvantaged tracts in Charlotte's urban core), and Guilford combines high equity burden with moderate utilization strain. Orange County ranks last (0.077), reflecting low Justice40 burden, adequate per-capita infrastructure, and a university-town demographic. Weight sensitivity analysis confirms the top-three set is stable across all tested scenarios. As equity weight rises, Mecklenburg gains and Union falls, which tracks with their different score compositions.

4.7.1 Observational, Not Experimental: OFAT Disclosure

The weight sensitivity analysis described above and in Section 6 perturbs the three pillar weights one factor at a time (OFAT) across the 0.30–0.50 equity-weight range while holding the utilization-to-cost-effectiveness ratio constant. This is computational sensitivity analysis, not a fractional-factorial or response-surface design. The "triply-robust" top-3 finding is that Union, Mecklenburg, and Guilford hold their positions across three independent perturbations: the cost-effectiveness sub-weight extremes documented in Section 4.7, the equity-weight range, and the remote-work multiplier values mathematically demonstrated to cancel in min-max normalization. This is robustness evidence under that restricted exploration, not a claim of design optimality across the full weight simplex. A full design-of-experiments treatment over the simplex is identified as future work. The analysis as conducted is observational rather than experimental:

counties are not randomly assigned to treatment conditions, demographic variables are not manipulated, and policy parameters are not set by the analyst. The framework is computational sensitivity analysis informed by DOE principles, applied to a fixed observational dataset.

4.8 Tools and Environment

All analysis was performed in Python, primarily in modular scripts organized by pipeline stage. GeoPandas and Shapely handled spatial operations including the tract-to-ZCTA crosswalk, station geocoding, and choropleth generation. Pandas and NumPy supported the tabular data pipeline across all five phases. Statsmodels provided ARIMA modeling, stationarity testing, and diagnostic checks for the time series forecasting. Matplotlib produced the publication-quality figures at 600 DPI with PDF export. The demand-side forecasting used SAS Model Studio, which auto-selected model types per county and generated the predictions validated in Phase 1.

Time series methods, particularly stationarity testing and out-of-sample validation design, informed the forecasting approach. A Variance Inflation Factor (VIF) check on the three scoring pillars confirmed multicollinearity is well within acceptable bounds: maximum pillar VIF of 1.41 (Equity), with Utilization at 1.08 and Cost-Effectiveness at 1.32, all well below the conventional 5.0 concern threshold. The detailed diagnostic results (VIF, Chow test, CI coverage decomposition, estimator method comparison) are consolidated in §4.9 below.

4.9 Diagnostic Suite

A quantitative reviewer reading the methodology benefits from seeing the full diagnostic battery in one place rather than scattered across §4.2, §4.4, §4.8, and §6.1. This subsection consolidates four diagnostics applied to the scoring framework and the underlying forecast inputs.

Multicollinearity (Variance Inflation Factor). A VIF check on the three scoring pillars produced maximum VIF 1.41 (Equity), with Utilization at 1.08 and Cost-Effectiveness at 1.32, all well below the conventional 5.0 concern threshold. Pairwise Pearson correlations confirm the same: Equity ↔ Utilization $r = -0.26$, Equity ↔ Cost-Effectiveness $r = +0.48$, Utilization ↔ Cost-Effectiveness $r = -0.03$. The moderate Equity ↔ Cost-Effectiveness correlation (+0.48) is substantively expected (disadvantaged counties also tend to have workforce demand) but does not approach a level that would inflate VIF. The three pillars are statistically independent dimensions, and the 0.40 / 0.35 / 0.25 weights reflect three distinct constructs rather than reshuffled copies of one underlying factor.

Structural-break testing (Chow). A Chow test applied to the 82-month NCDOT BEV registration series identified the August 2022 Inflation Reduction Act passage as a strongly significant structural break ($F = 1{,}268.35$, $p < 1 \times 10^{-6}$). The May 2025 NCDOT methodology change was also tested, but the result is degenerate ($n_2 = k = 2$: only two post-break observations within the training window, equal to the number of model parameters, producing a mechanically inflated F-statistic that cannot be interpreted). The May 2025 degeneracy is itself informative: it indicates the test window does not yet support inference about that break, and it identifies a future-work re-run when 12 or more months of post-methodology data exist (by mid-2026). The IRA break is the dominant structural feature in the series; the May 2025 methodology change is a minor perturbation (empirical magnitude approximately 0.4% of statewide totals, contributing roughly 20% of the observed underprediction bias) that works *against* the underprediction direction, making the validation a stricter test than the prior methodology would have produced.

Estimator method comparison (CSS versus MLE). SAS Model Studio uses Conditional Sum of Squares (CSS) as its default ARIMA estimator; Python statsmodels uses exact Maximum Likelihood Estimation (MLE) by default. A side-by-side comparison on a statewide ARIMA(1,1,1) produced forecast differences of less than 0.1% across all four-month-ahead horizons (23 to 104 vehicles on a base of approximately 97,000). The AR parameter differed by 0.06%, the MA parameter by 1.48%, and $sigma^2$ by 20% (affecting confidence interval width, not point forecasts). The estimator method choice does not materially affect any result reported in this study.

Confidence interval coverage decomposition. The forecast 95% prediction intervals achieved 75.50% raw coverage on the four-month holdout, below the nominal target. Decomposition explains the gap: observed actuals exceeded the upper bound in 18.0% of observations and fell below the lower bound in only 6.5%, an asymmetric 3:1 ratio that indicates the gap is a *centering* (bias) problem rather than a *width* problem. After county-specific bias correction (effectively shifting each county's forecast center by its measured bias), CI coverage recovers to 93.75%, essentially at the 95% nominal target. The widths of the prediction intervals are correctly calibrated; the centers are systematically low because BEV adoption accelerated faster than the pre-IRA training data anticipated. The correct policy response is not to widen the intervals but to apply the documented +4 to +5% upward buffer in forecast-based infrastructure allocations (see §4.7's forecast buffer sub-metric).

5. Data Collection

5.1 NCDOT BEV Registration Data

The demand side of the analysis draws on the complete battery electric vehicle registration time series published by the North Carolina Department of Transportation. NCDOT releases monthly Excel workbooks through its climate change documents page, each containing county-level counts for BEV, PHEV, hybrid, gasoline, and diesel registrations across all 100 North Carolina counties. A custom Python download utility (`ncdot_ev_pipeline.py`) retrieves these files by constructing URLs from year-month patterns and consolidates them into a single CSV using standardized column names and date formatting.

The baseline dataset spans September 2018 through June 2025, producing 8,200 county-month observations (100 counties across 82 months). The validation extension file covers all of calendar year 2025 (January through October), which contains 1,000 county-month records total, but January through June 2025 already appears in the baseline. The net new observations are the four-month holdout period, July through October 2025: 400 county-month records (4 months across 100 counties), bringing the deduplicated total to 8,600. Every county is present in every month; the dataset has zero missing values and statewide BEV totals show monotonic growth across the full time series.

One methodological discontinuity requires flagging. NCDOT changed its registration counting methodology after May 2025: the new methodology deduplicates registration records and excludes vehicles whose plates have been recently removed from circulation (documented on the NCDOT ZEV Registration Data page). The data structure remained identical, so the existing pipeline handled it without modification. A binary flag (`Methodology_PostMay2025`) marks post-change records. Empirical analysis places the magnitude of the change at approximately 0.4% of statewide totals (roughly 370 vehicles across 100 counties), with only 10 counties showing any month-over-month decrease at May 2025 and none exceeding 5 vehicles. The direction of the change *lowers* counts (fewer duplicates, fewer inactive vehicles), which means it works *against* the underprediction direction observed in the forecast validation; validation against this cleaner, lower baseline is therefore a stricter test than the prior methodology would have produced. The validation analysis described in Section 4.2 achieved a MAPE of 4.34% against this post-change period, supporting the conclusion that the discontinuity did not meaningfully disrupt the time series patterns the forecasting models rely on.

The BEV column is the primary analysis variable throughout the project. PHEV registrations are retained for market share calculations but are excluded from demand-side modeling because the NEVI program targets public charging infrastructure, and PHEVs can operate on gasoline when charging is unavailable. All counts are cumulative registrations, not new registrations per month.

5.2 AFDC Infrastructure Data

Infrastructure supply data comes from the U.S. Department of Energy's Alternative Fuels Data Center. The baseline analysis from Fall 2025 used a manually exported CSV filtered to DC fast chargers only: 1,725 connector-level records across 355 unique stations, with 74 columns covering station attributes, connector types, power output, network provider, access type, and operational status. That extract captured nearly all fast-charging ports (1,742 of 1,745 statewide) but missed the 82.4% of stations that offer only Level 2 charging, along with all 35 Level 1 stations.

A programmatic pull from the NREL AFDC API replaced that extract, downloading all operational electric stations in North Carolina regardless of charging level or access type. The

API script (`afdc_api_download.py`) queries the AFDC station locator endpoint with state, fuel type, and status parameters, returning the full station inventory. The February 2026 pull captured 1,985 stations with 6,145 total connectors across 267 cities and 358 ZIP codes. Access types break down to 1,876 public and 109 private stations. Charging level distribution is heavily weighted toward Level 2: 1,636 stations (82.4%) are L2-only, 311 are DCFC-only, 29 are mixed L2/DCFC, and 9 are L1-only. ChargePoint operates the largest share at 51.8% (1,028 stations). Connector types include J1772 (1,586), J1772COMBO (243), Tesla/NACS (234), and CHAdeMO (143), with power output ranging from 2 to 400 kW across the 1,611 stations where power data is available. All subsequent analysis uses the February 2026 API pull exclusively; the old DCFC-only file remains in the repository for reference but is not loaded by any current script.

Data cleaning removed 2 non-NC stations (ZIP codes 20020 and 39817), reducing the working dataset to 1,983 NC stations. Fifty-six co-located stations sharing exact coordinates were confirmed as legitimate separate equipment and retained. Join-key validation matched 354 of 358 AFDC ZIP codes to Census ZCTAs (98.9% match rate). The 4 unmatched ZIPs include 2 non-NC stations (removed in cleaning) and 2 PO Box ZCTAs (28608 in Boone and 27706 in Durham) with no Census population record; the 3 stations in these PO Box ZCTAs (one ZCTA hosting two stations) were excluded from population-based density calculations. High-missingness fields (EV pricing 86%, facility type 71.5%, owner type 70%) were documented and excluded from density analysis, while core location and port count fields are nearly complete at 99.8% coverage.

5.3 Census Data

Three Census Bureau products supply population, income, and geographic boundary data at different scales.

American Community Survey (ACS) 2022 five-year estimates provide household income distributions at the tract level through table B19001, which reports counts across 16 income bins. The tract-level income data feeds the workplace charging demand pipeline described in Section 4.6, where it bridges the gap between LODES individual earnings categories and the household income thresholds relevant to EV affordability. ACS also supplies housing tenure data (table B25003) used descriptively in the equity analysis. Statewide, 44.4% of North Carolina households report income above $75,000, and the renter share is 33.8%.

ZCTA population estimates, also from ACS 2022, cover 853 North Carolina ZCTAs with a total population of 10,470,214, representing 98.8% of the state's approximately 10.6 million residents. The distribution is right-skewed (mean 12,275 versus median 5,380, skewness 1.70), with 19 uninhabited ZCTAs excluded from per-capita calculations. Population data enters the analysis as the denominator in charging port density metrics (ports per 10,000 residents) and as the weighting factor in population-weighted Gini calculations.

County and ZCTA boundary shapefiles come from the Census TIGER/GENZ 2020 product at 500k resolution. All 100 county geometries are valid and use the NAD83 coordinate reference system (EPSG:4269). The 853 NC-prefix ZCTAs were filtered from a national ZCTA file using a spatial join against the state bounding box. Census tract boundaries in the 2010 vintage were required separately for the CEJST equity crosswalk, since CEJST v2.0 was released using 2010 tract definitions and no successor migration has occurred (see Section 5.5).

5.4 LEHD LODES Commuter Data

Workplace charging demand estimation uses the Census Bureau's Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) dataset, version 2021. LODES provides origin-destination commuter flows at the Census block level, the finest geography available in any free commuting dataset. The North Carolina origin-destination main file contains 3,768,428 intra-state commuter flow records with zero null values and zero duplicate block pairs. A companion Workplace Area Characteristics (WAC) file contains 71,921 workplace blocks.

LODES segments workers into three earnings categories: SE01 (below $1,250/month), SE02 ($1,251 to $3,333/month), and SE03 (above $3,333/month, roughly $40,000 annually). The SE03 category serves as the initial filter for EV-affordable commuters, reducing the pool from 4,198,163 total workers to 2,041,731. Additive decompositions (SA01+SA02+SA03=S000, SE01+SE02+SE03=S000, SI01+SI02+SI03=S000) verified for all records in the origin-destination file, and CE01+CE02+CE03=C000 verified for all WAC records. A Census block-to-county crosswalk achieved 100% coverage: every origin-destination and WAC block maps to a county, and all 100 North Carolina counties are present.

Alternative data sources were considered before LODES was selected. The Census Transportation Planning Products (CTPP) dataset, originally planned, was rejected for its older vintage (2012-2016 survey responses versus LODES 2021), county-level geography (versus LODES block-level), and restricted access. Commercial alternatives from StreetLight Data and Replica offered real-time mobility data but at costs ranging from $10,000 to $100,000, well outside a research-budget scope. The National Household Travel Survey provides only state-level commuting estimates, too coarse for county-level workplace demand.

5.5 CEJST / Justice40 Data

The Climate and Economic Justice Screening Tool version 2.0 (CEJST v2.0) was the federally-operational instrument for identifying disadvantaged communities under the Justice40 Initiative from its release in December 2024 through January 22, 2025. Developed by the Council on Environmental Quality (CEQ) and accessible at `screeningtool.geoplatform.gov`, CEJST v2.0 served as the federal screening standard during the period in which Executive Order 14008 (Justice40) was in force. The screening methodology, which categorizes a Census tract as 'disadvantaged' if it is both low-income (at or above the 65th percentile for households below 200% of the federal poverty level) and overburdened on at least one of eight environmental, health, or infrastructure dimensions, is documented in the CEJST v2.0 Technical Support Document (Council on Environmental Quality, 2024). That methodology document remains the federal record of the screening criteria, independent of the operational status of the screening tool itself.

The dataset covers 2,195 North Carolina Census tracts at 2010 tract boundaries, of which 934 of 2,170 evaluable tracts (43.0%) carry the disadvantaged designation after excluding 25 zero-population tracts (12 water-body, 13 institutional and military). The 43.0% rate substantially exceeds the national average of approximately 37%, reflecting NC's mix of rural poverty in eastern NC and Appalachia, above-average health burdens, housing cost stress, and legacy industrial contamination.

The CEJST tool was removed from `screeningtool.geoplatform.gov` on January 22, 2025 following the rescission of Executive Order 14008. The EDGI / Public Environmental Data Partners (PEDP) coalition archived and restored the data within days. This study uses the community-preserved CEJST v2.0 (December 2024 release) downloaded as a bulk CSV from the PEDP mirror. Data integrity was verified against pre-removal records. The CEQ Technical

Support Document, the federal record of the screening methodology, was preserved independently and remains accessible. The archival pathway is therefore a *data continuity* matter rather than a methodological one: the methodology is fully reproducible from the federal record, and the data itself has been preserved by an independent research coalition with chain-of-custody documentation. No API exists for CEJST; bulk CSV is the only access path under either the original government distribution or the community archive.

No federal successor instrument exists at the time of this writing. The EPA EJScreen tool, which is sometimes proposed as a substitute, uses different burden categories and different threshold percentiles and is not a drop-in replacement for CEJST v2.0. Future federal action may produce a replacement; this study's CEJST-anchored equity findings should be re-examined against any such successor when it becomes available, with the understanding that the underlying disadvantaged-community designations may shift. Section 3.4 discusses the broader policy context.

5.6 Data Quality and Limitations

The most consequential limitation across the data collection is temporal mismatch. The NCDOT registration series runs through October 2025. The AFDC infrastructure pull reflects station status as of February 2026. ACS income and population estimates use 2022 five-year averages (representing 2018–2022 survey responses). LODES reports 2021 commuting patterns. CEJST uses indicators drawn from multiple federal datasets with varying reference periods. No two sources share the same temporal window, and the analysis cannot synchronize them.

The LODES vintage carries a specific concern. The 2021 data captures commuting patterns during a period when COVID-era remote work was still elevated relative to pre-pandemic norms. The 0.85 remote work multiplier applied in the demand pipeline (Section 4.6) adjusts for this, but it is a national average from Barrero, Bloom, and Davis (2023), not a North Carolina-specific estimate. Sector-specific and geography-specific rates could differ. The project brackets this uncertainty by reporting results under low, baseline, and high scenarios.

CEJST's use of 2010 Census tract boundaries against a study built on 2020-vintage ZCTAs and county boundaries introduces a crosswalk requirement. At the county level the mismatch is negligible because North Carolina's county boundaries are geographically identical between the two Census vintages. At the sub-county level, the area-weighted interpolation used in the tract-to-ZCTA crosswalk (Section 4.5) assumes uniform population distribution within tracts, which introduces modest error in large, heterogeneous rural tracts.

One data gap shaped the project's analytical scope from the outset: NCDOT does not publish ZIP-code-level BEV registration counts. The ZIP-code infrastructure density analysis described in Section 4.4 measures supply-side inequality using port density per 10,000 residents, but it cannot compute BEV-to-port ratios at the ZIP level because the demand denominator does not exist at that geography. Commercial alternatives (IHS Markit, Experian VIO) offer ZIP-level vehicle registration data but require licenses in the range of $5,000 to $20,000 per state. The analysis works around this by using county-level BEV counts for demand and ZIP-level port counts for supply inequality measurement, accepting that the two scales do not perfectly align.

Finally, the AFDC data has high missingness on several descriptive fields: EV pricing information is absent for 86% of stations, facility type for 71.5%, and owner type for 70%. These fields were excluded from the density analysis. Core location and port count fields are nearly complete (99.8% of stations matched to Census ZCTAs), and the two non-NC stations that appeared in the dataset (ZIP codes 20020 and 39817) were removed during cleaning.

These limitations are taken up systematically in Section 9 (Limitations), where they are placed alongside three additional federal-policy-specific limitations (CEJST methodology lock at v2.0, NEVI program rule evolution risk, and state-administered formula program variance) and the direction-of-bias analysis that accompanies each one.

6. Results

6.1 Forecast Validation

The out-of-sample validation tested SAS Model Studio's county-level BEV adoption forecasts against four months of data the models had never seen (July through October 2025). The mean absolute percentage error across all 100 counties was **4.34%**, below the 5% usability threshold established in the project's success criteria. Three model families contributed: exponential smoothing (82 counties), ARIMA (13), and unobserved component models (5). None required retraining for the holdout period. Figure 44 plots actual versus predicted registrations across all 400 county-month observations; points above the 45-degree line indicate the systematic underprediction pattern discussed below.

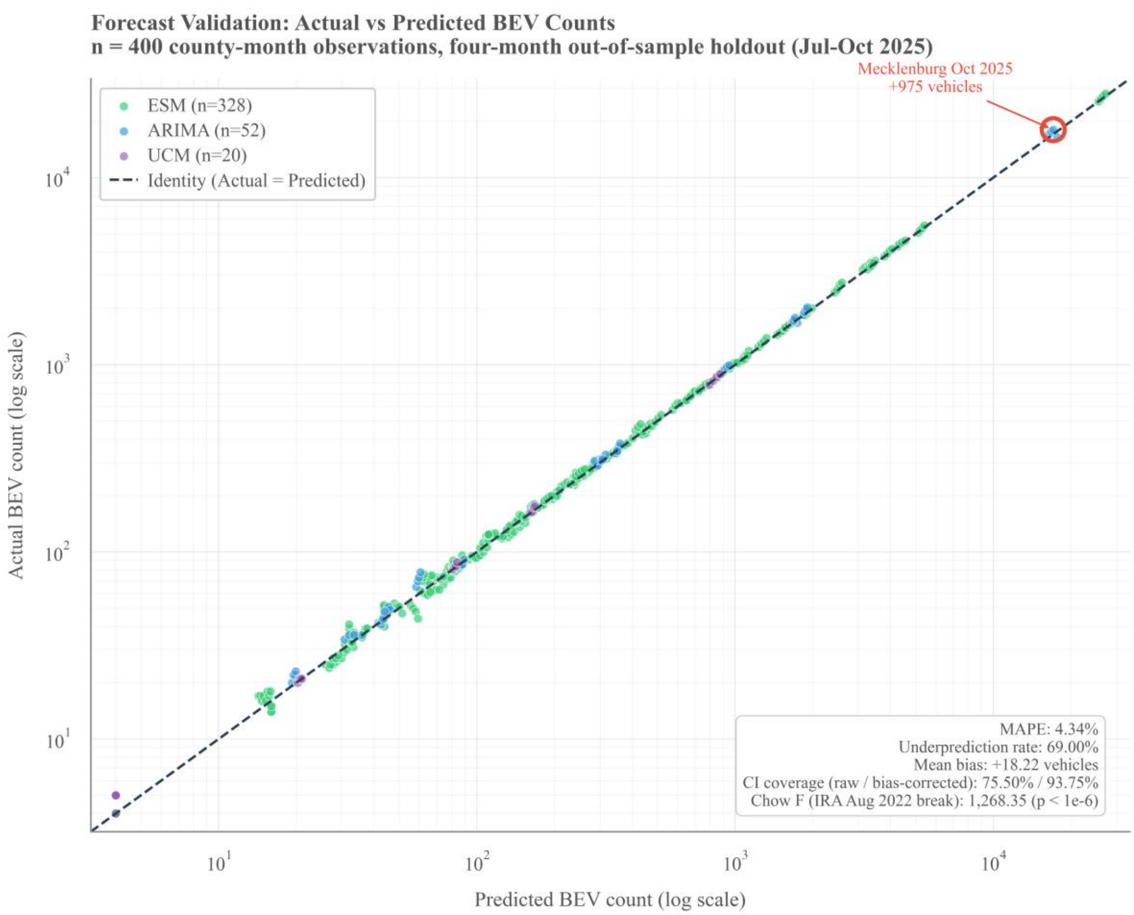

Figure 44

Forecast Validation: Actual Versus Predicted BEV Registrations Across 400 County-Month Observations

The accuracy figure, however, tells only part of the story. Actual registrations exceeded forecasts in **69.00%** of observations, with a mean bias of **+18.22** vehicles per county-month. The 95% prediction intervals achieved **75.50%** raw coverage on the holdout, well below the nominal 95%. As decomposed in §4.9, this gap reflects a *centering* (bias) problem rather than a *width* problem: after county-specific bias correction, coverage recovers to **93.75%**, essentially at the nominal target. A Chow structural-break test on the 82-month registration series confirms that the underprediction reflects a real adoption acceleration: F = **1,268.35** ($p < 1 \times 10^{-6}$) at the August 2022 Inflation Reduction Act passage date. The models learned historical growth patterns that are now too conservative. Mecklenburg County in October 2025 illustrates the extreme: actual registrations exceeded the forecast by 975 vehicles in a single month.

The systematic underprediction has a concrete policy consequence. Infrastructure planning built on these forecasts without adjustment would undersize deployment by roughly 4 to 5%. The study carries this buffer forward into the scoring equation's utilization pillar, applied globally rather than per-county because county-specific bias estimates over a four-month window would overfit.

Table 1

Headline Findings

#	Finding	Value	Section
1	Out-of-sample forecast validation (MAPE, n = 400 county-months)	**4.34%**	§6.1
2	Forecast underprediction rate	**69.00%**	§6.1
3	IRA structural break (Chow F-statistic, $p < 1 \times 10^{-6}$)	**1,268.35**	§4.9

4	CI coverage (raw / county-specific bias-corrected)	**75.50% / 93.75%**	§4.9
5	Statewide BEV-to-port ratio (NC, Feb 2026; IEA global benchmark ≈ 10)	**16.9**	§1.2, §6.4
6	County-level BEV ownership Gini	**0.805**	§1.2, §6.3
7	ZIP-level infrastructure Gini (statewide, population-weighted)	**0.566**	§6.3
8	Theil-T within-county share of inequality (top-10 cohort)	**84.5%**	§6.3
9	Justice40-designated NC Census tracts (CEJST v2.0)	**43.0%** (934 of 2,170)	§3.4, §6.5
10	Stations sited in Justice40 tracts (top-10 county scope, as of Jan 21, 2025)	**24.5%** (296 of 1,210)	§6.5
11	Top-3 NEVI counties (stable across weight sensitivity)	Union (0.561), Mecklenburg (0.548), Guilford (0.465)	§6.6
12	Scoring pillar independence (maximum VIF; concern threshold 5.0)	**1.41**	§4.9

Note. The table consolidates the most-cited numerical findings of this study for readers who may engage at a skim level. Section pointers indicate where each finding is established.

6.2 Infrastructure Coverage Gaps

The February 2026 AFDC API pull revealed that the infrastructure picture is substantially different from what the baseline's DCFC-only extract showed. Of 853 Census ZCTAs in North Carolina, 499 (58.5%) have zero charging stations of any type. Those 499 ZCTAs represent 2.2 million people, or 21.1% of the state's population. The coverage gap is not concentrated in remote rural areas alone; 9 of the 10 largest counties by BEV count have at least one ZIP code in the top 20 most underserved list.

Within the top 10 counties, 134 ZIP codes serve 4.4 million residents with 3,722 charging ports. The mean density is 11.18 ports per 10,000 population, but the median drops to 5.51, dragged down by the long right tail of a few high-density commercial ZIPs. The spread between best- and worst-served ZIPs is enormous. Charlotte Uptown (28202) sits at 78.64 ports per 10,000 people. Fourteen miles east, Charlotte ZIP 28215 has 64,713 residents served by 2 ports, producing a density of 0.31. That is a 250-fold gap within the same county. Burlington (27215) fares even worse in absolute terms: 1 port for 45,982 people, a density of 0.22 per 10,000 that is roughly 50 times below the statewide median.

The top 20 underserved ZIPs collectively hold 732,892 residents served by 58 ports. Fifteen of those 20 ZIPs have zero DC fast chargers. Mecklenburg and Guilford each contribute 5 ZIPs to the top 20, confirming that the state's two most infrastructure-rich counties also contain some of its worst gaps. Several underserved ZIPs nominally have chargers that turn out to be private access only (Winston-Salem 27105, for instance), making their effective public supply zero.

6.3 Intra-County Inequality

The Gini coefficient, computed on ports per 10,000 population across populated ZCTAs within each county, produces a statewide population-weighted value of 0.566. By standard interpretation that qualifies as very high inequality. Mecklenburg tops the list at 0.623, followed by Guilford at 0.571. At the other end, Durham records the lowest Gini among the top 10 counties at 0.408, which is still high in absolute terms. Durham's weighted Gini (0.408) diverges meaningfully from its unweighted value (0.51), because the well-served ZIPs happen to be the populated ones. In Mecklenburg, no such divergence appears; large-population ZIPs are underserved alongside small ones.

Union County presents an interpretive challenge that is easy to misread. Its Gini is low, which might suggest equitable distribution, but the underlying reason is that every ZIP is equally underserved. The interquartile range of port density across Union ZIPs is just 0.92. Equality at a low level of service is not the same as adequate access.

The Gini captures the shape of the distribution but cannot decompose it spatially. The Theil-T (GE(1)) index, which is exactly decomposable into between-group and within-group components via the additive decomposability of the GE(1) class (Bourguignon, 1979; Shorrocks, 1980), answers the question the Gini leaves open. Total Theil-T for the top 10 counties is 0.5791. Of that total, 0.0900 (15.5%) falls between counties and 0.4892 (**84.5%**) falls within counties. A robustness check using Theil-L (GE(0)) confirmed the pattern at 82.5% within-county. Both decompositions verified to machine precision (difference of 2.22×10^{-16} between summed components and total). Figure 33 presents this decomposition as a stacked bar chart.

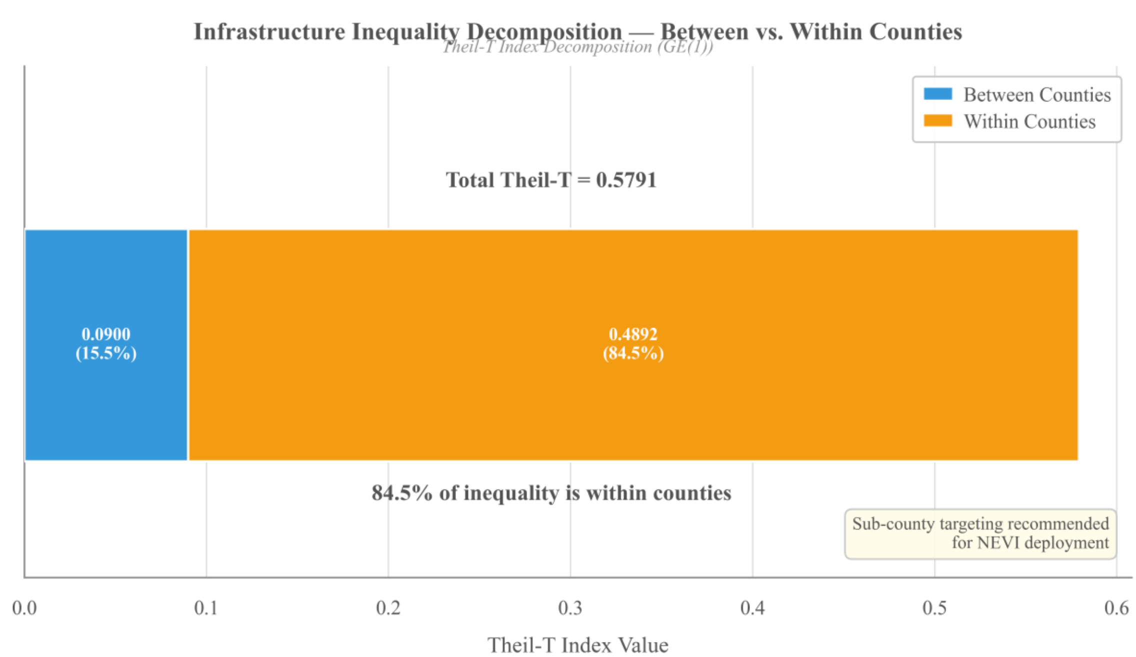

Figure 33

Theil-T Decomposition of EV Charging Infrastructure Inequality: Within-County vs Between-County

Mecklenburg alone contributes 41.5% of all within-county inequality. Wake adds 28.1%. Together the two counties account for approximately 70% of the within-county component, driven by the combination of large populations and extreme internal variation. Guilford, despite having the second-highest Gini, contributes only 7.7% of within-county inequality because its population is smaller. Union contributes 0.6%, consistent with its uniform-but-low service pattern.

The 84.5% within-county finding carries a direct policy implication. County-level allocation formulas, which is how most state NEVI programs distribute funding, would miss the majority of the inequality problem. Figure 24, the Mecklenburg heat map, makes this visible: Charlotte Uptown appears as a dark cluster of concentrated infrastructure surrounded by pale green and hatched zero-station areas that include 28215 and its 64,713 residents. That spatial pattern is what a Gini of 0.623 looks like on a map.

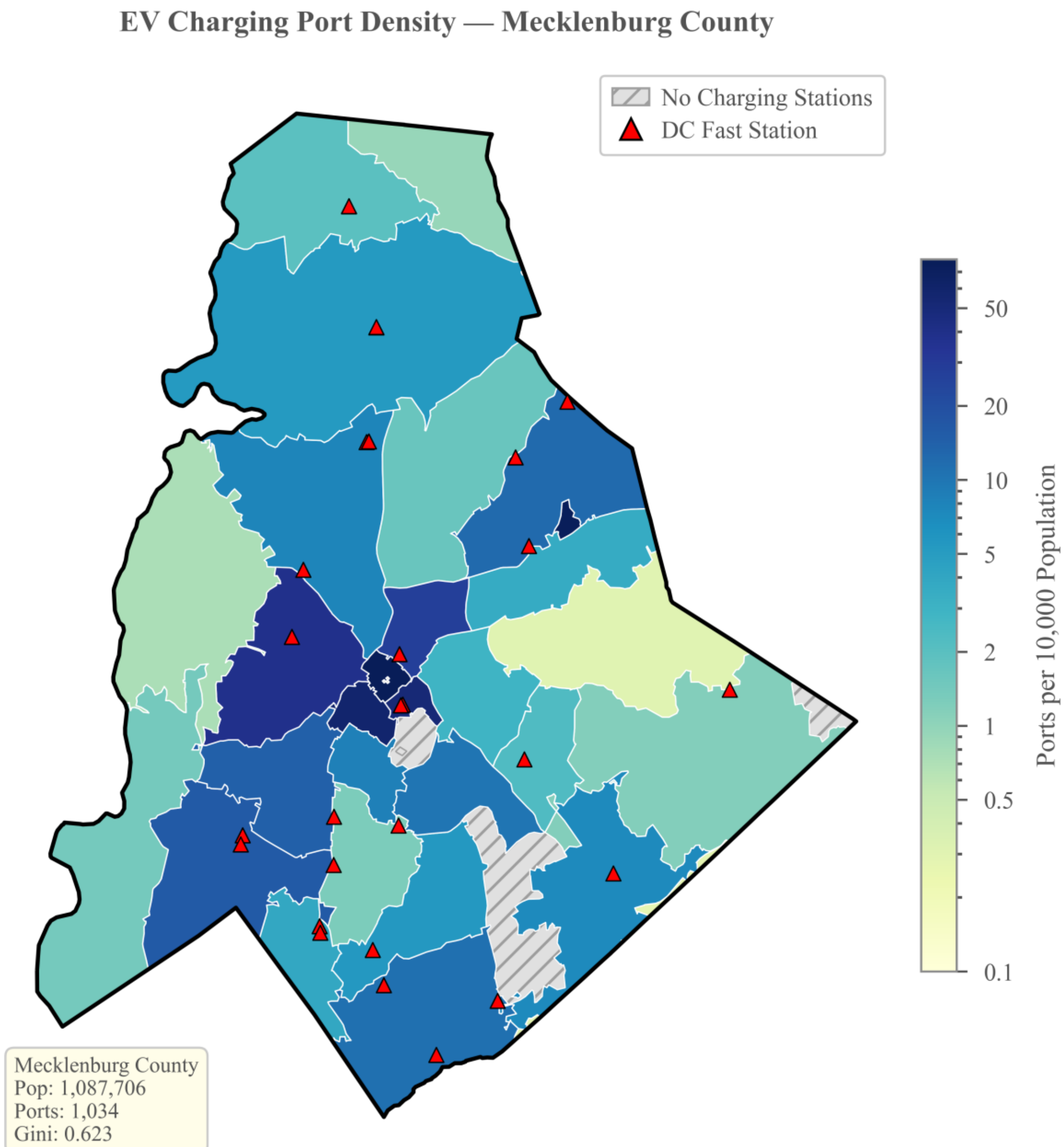

Figure 24

Mecklenburg County: Charging-Station Density Heat Map

6.4 Workplace Charging Demand

Layering commuter flows onto residential registration data changes the demand picture substantially. Mecklenburg County absorbs +194,361 net commuters per day, Wake +126,517, and Durham +89,450. After the three-layer adjustment pipeline (SE03 income filter, ACS income correction to the $75,000 household threshold, and 0.85 remote work reduction), statewide adjusted workplace demand totals 859,260 workers. The filtering proceeds in three steps: 4,198,163 raw workers to 2,041,731 after SE03 selection (48.6%), to 1,010,894 after the income correction (49.5% of SE03), to 859,260 after the remote work multiplier.

The county typology that emerges from net commuter flows classified all 100 NC counties into three groups: 8 employment centers (net inflow above +10,000), 75 balanced counties, and 17 bedroom communities (net outflow below -10,000). Union County, which ranks first in the NEVI scoring, is not an employment center; it is a bedroom community of Charlotte with a net outflow of -36,113 commuters per day. Its residents drive to Mecklenburg for work, which means Union's residential BEV-to-port ratio understates its charging need during the workday and Mecklenburg's understates its daytime demand from non-residents.

At baseline assumptions (30% charging adoption rate, 15:1 port ratio for workplace deployments), statewide workplace port need is 17,185 ports (13,748 Level 2, 3,437 DCFC). Mecklenburg alone needs 3,658 workplace ports, Wake 3,391, Durham 1,496. The 15:1 workplace planning ratio reflects the predictable mid-morning peak and 4–8 hour dwell time at employment centers, which enable more vehicles per port than typical public residential or destination charging deployments. For context, NC's statewide public BEV-to-port ratio is 16.9 (NCDOT registrations divided by AFDC ports as of February 2026), and the International Energy Agency's global benchmark for public chargers sits at approximately 10 BEVs per port (IEA Global EV Outlook, 2024). The workplace charging efficiency feeds the cost-effectiveness pillar of the scoring equation, where Mecklenburg scores highest (0.801) and Buncombe lowest (0.034), reflecting the difference between a 194,000-commuter employment center and a tourism-driven economy with lower commuter volumes. Figure 36 compares residential-only and residential-plus-workplace demand side by side for the 10 scoring counties.

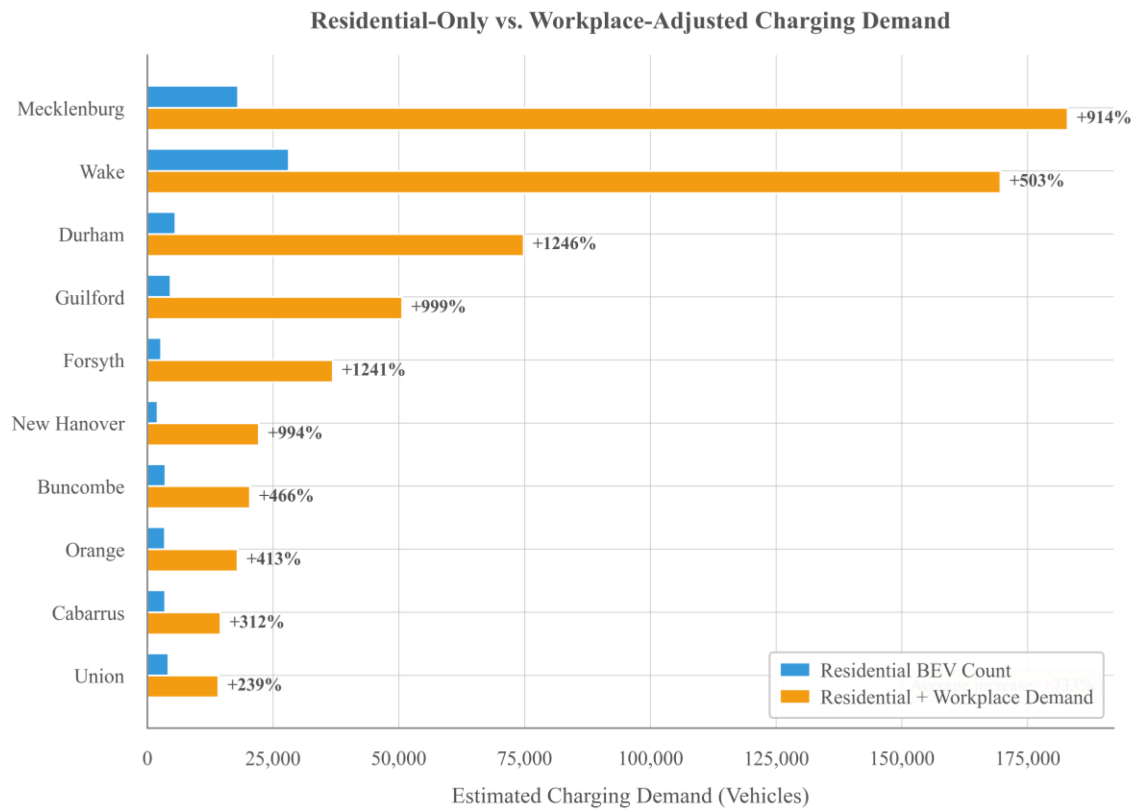

Figure 36

Workplace Charging Demand: LODES-Adjusted vs Unadjusted Estimates

6.5 Equity Analysis

CEJST v2.0 designates 934 of 2,170 evaluable North Carolina Census tracts (43.0%) as disadvantaged under the Justice40 definition, substantially above the national average of approximately 37% of tracts. The elevated rate reflects the state's combination of rural poverty in eastern NC and Appalachia, above-average health burdens (diabetes, heart disease), housing cost stress, and legacy industrial contamination. Within the project's top 10 BEV counties, the population-weighted disadvantaged rate drops to 18.5%, which is expected given that these are North Carolina's wealthiest urban and suburban counties. The county-level rates vary widely: Guilford at 29.2%, New Hanover at 28.8%, Durham at 26.6%, Mecklenburg at 23.7% (the largest absolute count), and Wake at only 8.1% despite having the highest BEV registrations.

The climate change category sensitivity analysis tested what happens when the most debatable CEJST component is removed entirely. Of 934 disadvantaged tracts, 124 (13.3%) are flagged solely by climate change indicators. Removing them drops the statewide rate from 43.0% to 37.3%, essentially matching the national average. But the impact on the study counties is minimal: 7 of 10 are completely unaffected, and the three affected counties (New Hanover at -4.7 percentage points, Durham at -3.4, Wake at -2.2) shift modestly. The equity rankings, in other words, are not artifacts of CEJST's climate projections; they are driven by empirical health, housing, and workforce indicators.

A spatial overlay of charging station locations on CEJST-designated disadvantaged tracts quantifies how the existing infrastructure aligns with the Justice40 communities. Of the 1,210 stations within the top-10 study county scope, **296 (24.5%) sit in census tracts designated disadvantaged under CEJST v2.0 as of January 21, 2025**, the last operational day of the federal screening tool before its removal. Anchoring the finding to a specific date converts a moving-target concern (the post-rescission CEJST landscape) into a fixed measurement against the federal record. The 24.5% station share sits 18.5 percentage points below the 43.0% statewide share of disadvantaged tracts, but a more directly comparable benchmark is the 18.5% population-weighted disadvantaged rate within the study counties themselves: against that comparator, the station share runs approximately proportional with a modest surplus (24.5% vs 18.5%) but with strong county-by-county variation. Some study counties site stations roughly in proportion to their disadvantaged-tract population; others show pronounced under-representation that drives the equity component of their NEVI scores.

Figure 42, which overlays existing charging station locations on Justice40-designated tracts, is the most immediately legible output of Phase 5. The visual disconnect between where stations cluster and where disadvantaged populations live requires no statistical interpretation.

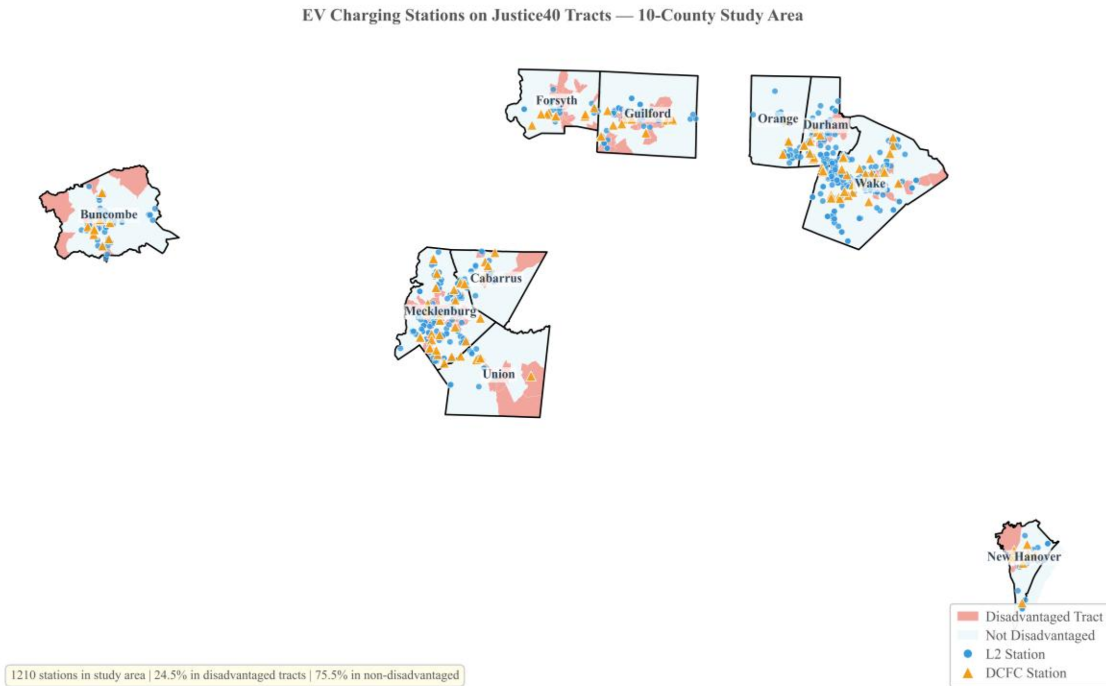

Figure 42

Charging Stations Overlaid on Justice40-Designated Disadvantaged Tracts

6.6 NEVI Priority Scores

The scoring equation described in Section 2.3 produces a single composite ranking for each of the 10 study counties. The full results, computed from `scoring_framework_final.py` and verified against the NEVI equation arithmetic to within 1×10^{-12} (Section 4.9), are shown in Table 2.

Table 2

NEVI Priority Scores for the Top 10 NC Counties by BEV Registration

Rank	County	NEVI Score	Equity	Utilization	Cost-Eff.
1	Union	0.561	0.319	1.000	0.333
2	Mecklenburg	0.548	0.810	0.067	0.801
3	Guilford	0.465	0.855	0.103	0.347

4	New Hanover	0.341	0.607	0.071	0.293
5	Wake	0.316	0.322	0.130	0.568
6	Durham	0.313	0.469	0.033	0.454
7	Forsyth	0.285	0.283	0.156	0.467
8	Cabarrus	0.199	0.228	0.229	0.111
9	Buncombe	0.197	0.470	0.000	0.034
10	Orange	0.077	0.059	0.101	0.071

The top three counties arrive at high scores through different pathways. Union is utilization-driven: its 101.5 BEV-per-port ratio is more than three times the next-highest county (Cabarrus at 32.0), producing a utilization score of 1.000. This is a suburban growth corridor where infrastructure deployment has not kept pace with adoption. Mecklenburg is equity-driven: its equity sub-score of 0.810 is the second-highest in the cohort (behind Guilford at 0.855), and its cost-effectiveness score of 0.801 reflects the scale of its employment center. Guilford combines the highest equity burden with moderate utilization strain, placing it third.

Wake County ranks fifth despite having the most BEV registrations in the state. The reason is legible in the component scores: its equity score of 0.322 reflects a relatively low Justice40 burden (only 8.1% of its tracts are disadvantaged), and while its cost-effectiveness is strong (0.568, second overall), the equity weight of 0.40 pulls the composite down. The scoring equation is doing what it was designed to do: weighting equity consistent with the Justice40-aligned federal benchmark rather than rewarding raw demand.

Orange County sits last at 0.077, scoring low across all three pillars. Its Justice40 population-weighted burden is 4.9%, its per-capita infrastructure is adequate, and its university-town demographic produces a small EV fleet that is well served by existing stations.

Weight sensitivity analysis tested equity weights from 0.30 to 0.50 in 0.05 increments while holding the utilization-to-cost-effectiveness ratio constant at 7:5. The top-three set (Union, Mecklenburg, Guilford) remained unchanged across all five scenarios. The internal ordering shifted in predictable ways: as equity weight rises, Mecklenburg gains and Union falls, consistent with their different score compositions. Orange held steady at tenth in every scenario. The rankings are not artifacts of a particular weighting choice (see Section 2.3 for the weight rationale; combined with the cost-effectiveness sub-weight robustness and the remote-work multiplier mathematical cancellation in min-max normalization, this is the "triply-robust" top-3 finding referenced in §4.7.1).

6.7 NEVI Priority Score Visualization

Figure 43 visualizes the rankings in Table 2 as a horizontal stacked bar chart, with each county's total NEVI Priority Score decomposed into its three pillar contributions (Equity at 0.40, Utilization at 0.35, Cost-Effectiveness at 0.25). The figure renders the three archetype patterns immediately legible: Union's bar shows a dominant Utilization component (the 1.000 normalized utilization score driving the #1 ranking despite moderate equity), Mecklenburg's bar shows roughly balanced Equity and Cost-Effectiveness contributions (the equity-driven archetype), Guilford's bar shows a dominant Equity component (the highest equity sub-score in the cohort at 0.855), and Orange's short bar reflects low scores across all three pillars. The figure uses the Okabe-Ito colorblind-safe palette (Equity = #0072B2, Utilization = #E69F00, Cost-Effectiveness = #009E73), sorts counties by total NEVI score in descending order, and annotates each bar with the total score to three decimal places matching Table 2. Three archetype callouts identify Union as utilization-driven, Mecklenburg as equity-driven, and Orange as low-across-pillars. The figure is rendered at 600 DPI for publication output.

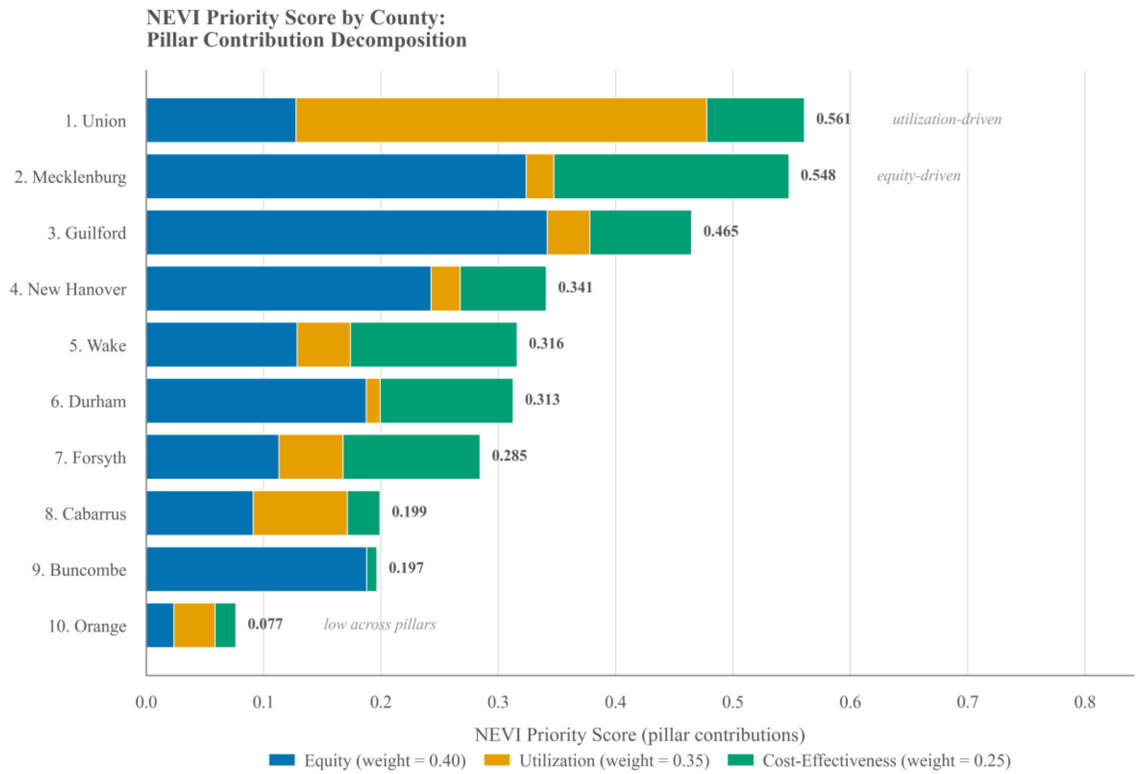

Figure 43

NEVI Priority Scores: Top-10 NC Counties

7. Discussion

This section synthesizes the analytical findings across the five phases of the study and the methodology review conducted alongside them. The synthesis is organized around three frames. The first is a cross-lane methodology synthesis: how the data engineering, spatial geometry, federal compliance, and econometric layers each substantiate a specific piece of the NEVI Priority Score's defensibility. The second is a stakeholder-by-stakeholder mapping of what the study delivers to each of the constituencies the framework is meant to serve. The third is the three-archetype interpretation of the top-10 county rankings, which converts an ordered list into a policy conversation by naming *why* each county sits where it does. The discussion below shows how each of the three frames contributes to the study's goal of supplying a prescriptive layer that makes findings actionable for policymakers.

7.1 Cross-Lane Methodology Synthesis

The defensibility of the NEVI Priority Score does not rest on any single methodological choice. Four distinct lanes of the data quality review (data engineering, spatial geometry, federal compliance, and econometrics) each contribute a specific substantiation that the rankings reported in Section 6.6 (and visualized in Figure 43) are not opinions but numbers a federal auditor can defend. The four substantiations are independent of each other and converge on the same conclusion, which is what gives the framework its robustness.

From the data engineering perspective, the strongest finding is that the CEJST tract-to-ZCTA crosswalk validation suite (12 checks in 3 tiers, 23 of 23 sub-checks passing at the committed analysis state) is the best-validated component in the analytical pipeline. The component that touches Justice40 compliance, which is the highest-sensitivity federal lane the study interacts with, is also the component most rigorously checked for geometric and statistical integrity. This is the reverse of the usual data-quality risk pattern, where the most consequential component is also the weakest. The 0.40 equity weight rests on geometry that has been formally tested to federal precision (Section 4.5), not on unvalidated overlays. The broader governance surface is audit-ready in a similar way: a consolidated data dictionary covering all six datasets, a 19-entry provenance table, a reproducibility lockfile (`uv.lock`), and 27 analysis scripts with documented transforms. The internal assertion battery in `scoring_framework_final.py` would fail the run before producing a corrupt CSV. The geometry holds.

From the spatial perspective, every area-dependent operation in the pipeline uses NC State Plane (EPSG:32119, meters); every topological-only operation uses WGS84 (EPSG:4326) with documented reasoning. No area or length calculation in the analytical pipeline happens in a geographic coordinate reference system. The 2010-to-2020 tract-vintage mismatch, which could have been a deal-breaker for a Justice40 auditor, is immaterial at the county level because North Carolina's county boundaries are geographically identical between the two Census decades, and the crosswalk's per-ZCTA area-conservation check (within 1%) backstops the claim empirically. The overlays that decide which ZCTAs carry which `justice40_pct`, and therefore which counties rank high on the 0.40 equity pillar, are not approximations of sound; they are sound.

From the federal-compliance perspective, the Data Governance Readiness chain documented in Section 5.6 and the consolidated provenance in Section 5 establish an end-to-end traceability path that a reviewer can actually traverse. Raw data immutable; acquisition scripts in `src/data-acquisition/`; provenance table; consolidated data dictionary; 27 analysis scripts; validation suites (MAPE 4.34%, crosswalk 23/23, Phase 4 EDA 59/59); scored outputs; reproducibility lockfile. The CEJST archival pathway is disclosed honestly and at appropriate prominence (Section 5.5): the federal screening tool was decommissioned January 22, 2025; the community archive preserves the data; the CEQ Technical Support Document preserves the methodology; no federal successor instrument exists. The directional analysis of three documented limitations follows the same pattern. The NCDOT methodology change works *against* the underprediction (Section 5.1); interstate commuter exclusion *undercounts* workplace demand (a floor, not a ceiling); and the remote-work multiplier *cancels* mathematically in normalization (Section 4.7.1). Each potential reviewer challenge becomes a disclosure that protects the analysis rather than wounding it.

From the econometric perspective, three headline numbers carry the statistical backbone of the entire scoring framework. The MAPE of 4.34% on a true four-month out-of-sample holdout (Section 6.1) means the forecasts are accurate enough for county-level capital allocation decisions. The maximum Variance Inflation Factor of 1.41 across the three scoring pillars (Section 4.9) means the 0.40 / 0.35 / 0.25 weights reflect three statistically independent dimensions of need, not reshuffled copies of the same underlying factor. The Chow F-statistic of 1,268.35 at the August 2022 IRA passage date (Section 4.9) converts the 69.00% underprediction from a model failure into the statistical fingerprint of an accelerating adoption regime. The infrastructure gap the study identifies is a *floor*, not a ceiling. The triply-robust top-3 finding (Section 4.7.1) shows Union, Mecklenburg, and Guilford holding their positions across the equity-weight sensitivity sweep, the cost-effectiveness sub-weight robustness, and the mathematically demonstrated cancellation of the remote-work multiplier. Combined with the diagnostics above, the rankings are not opinions but a structurally stable result.

7.2 Stakeholder Implications

The five expert lanes of the methodology review resolve into a mapping between the integrated finding and each stakeholder group identified in the project's stakeholder value analysis. What each group walks away with is summarized below.

NCDOT and FHWA (NEVI administrators). What the administrators take from this study is a ranked list of 10 counties (Union, Mecklenburg, Guilford, New Hanover, Wake, Durham, Forsyth, Cabarrus, Buncombe, Orange) whose top-3 ordering survived every robustness test the methodology review could apply, whose scoring weights are multicollinearity-defensible (VIF 1.41), whose underlying forecasts validated at 4.34% MAPE on a true out-of-sample holdout, and whose equity pillar geometry passes a 23-of-23 federal-standard validation suite. The list covers 10 counties that hold 73% of NC's BEV fleet; statewide extension is engineered into the framework and positioned as future work (Section 11). The deliverable is a priority ranking, not a dollar allocation. Dollar-level allocation would require per-station cost modeling and deployment-timeline assumptions that are outside this study's scope.

Federal Justice40 auditors and OIG-style reviewers. A federal auditor encounters several substantiating artifacts in this study. The crosswalk methodology aligns to EPA EJScreen and HUD USPS Crosswalk standards (Section 4.5), and the 12-check 3-tier validation suite passes 23 of 23 sub-checks. The CEJST v2.0 archival pathway is disclosed with verified schema match and a CEQ Technical Support Document anchor (Section 5.5). A climate sensitivity analysis shows 7 of 10 study counties unchanged even if the most-contested CEJST category is removed entirely (Section 4.5). The Theil-T decomposition (84.5% within-county inequality; Section 6.3) reframes the equity problem as intra-urban, which is where the study's sub-county analysis adds the most value. The 0.40 equity weight is not a guess; it is statistically independent of the other two pillars (Section 4.9) and reflects the Justice40 benefit-flow target established under Executive Order 14008, treated as an analytical anchor rather than a current federal mandate given the post-rescission landscape (Section 3.4).

County and regional planners. For planners, the most actionable output is the sub-county (ZIP-level) infrastructure-gap analysis, which is usable for local site-selection conversations. The 250-fold density gap between Charlotte ZIP 28202 (78.64 ports per 10,000 residents) and Charlotte ZIP 28215 (0.31 ports per 10,000), both in Mecklenburg County 14 miles apart, is the operational form of the Theil decomposition finding (Section 6.3). The ZIP-level outputs in `data/processed/phase3-*.csv` are the data planners can feed into grant applications and local prioritization processes. The three county archetypes (Section 7.3 below) tell planners which kind of intervention is appropriate for their county type: Union-type counties need *more stations*, Mecklenburg-type counties need *better-targeted stations*, and Orange-type counties are already relatively well-served. For planners outside the top-10 counties, the framework is designed to rank them too, pending the statewide extension positioned as future work.

Private-sector charging-network operators. Operators receive a competitive baseline anchored on a federal infrastructure inventory: the February 2026 AFDC API pull, 1,985 stations with 6,145 connectors. The data produces a market-gap number (NC statewide 16.9 BEVs per public port versus the IEA Global EV Outlook 2024 benchmark of approximately 10), a ranked set of counties where demand is under-supplied relative to BEV fleet size, and a workplace-charging market sizing from LEHD LODES commuter flows (859,260 adjusted commuter demand statewide). The white-space identification is not a vague claim; it is traceable to specific ZIPs with specific density ratios and specific demographic profiles.

Academic and research community. Researchers can use this study as a reproducible methodology backed by a Claims-to-Evidence Matrix that ties every numerical assertion to a source file (Appendix A) and a dependency lockfile that supports re-execution. The methodological contribution is the first additive Theil-T decomposition of EV charging infrastructure inequality applied to a state-level allocation problem (per the literature check in Section 3.3). Two supporting contributions follow: a tract-to-ZCTA area-weighted crosswalk that integrates federal equity data with local planning units (Section 4.5), and a climate sensitivity analysis demonstrating CEJST robustness for the NC context (Section 4.5). All five Unique Contributions identified in Section 10 are substantiated by the methodology review process, and the review methodology itself is documented as a contribution in its own right (Appendix B).

Public and Justice40 communities. For the publics affected by NEVI deployment, the study offers procedural and distributional fairness: the allocation methodology is transparent, the scoring formula is published, the equity pillar carries the largest weight (0.40), the geographic scope is disclosed (73% of NC BEV fleet, top-10 counties by BEV registration), and the underlying data sources are all public so the analysis is reproducible by anyone. The audit-ready framing is operational rather than rhetorical: every numerical claim in this manuscript is traceable to a source file via the Claims-to-Evidence Matrix in Appendix A and a public GitHub repository.

7.3 Three County Archetypes

The top-10 ranking in Section 6.6 sorts counties by composite NEVI Priority Score, but the ordered list alone does not tell NCDOT *why* each county sits where it does. The three-archetype framing converts the ranking into a policy conversation. Figure 45 visualizes the framing as a two-dimensional scatter in (Equity × Utilization) pillar space, with bubble size proportional to the composite NEVI score and quadrant background shading that names each archetype directly.

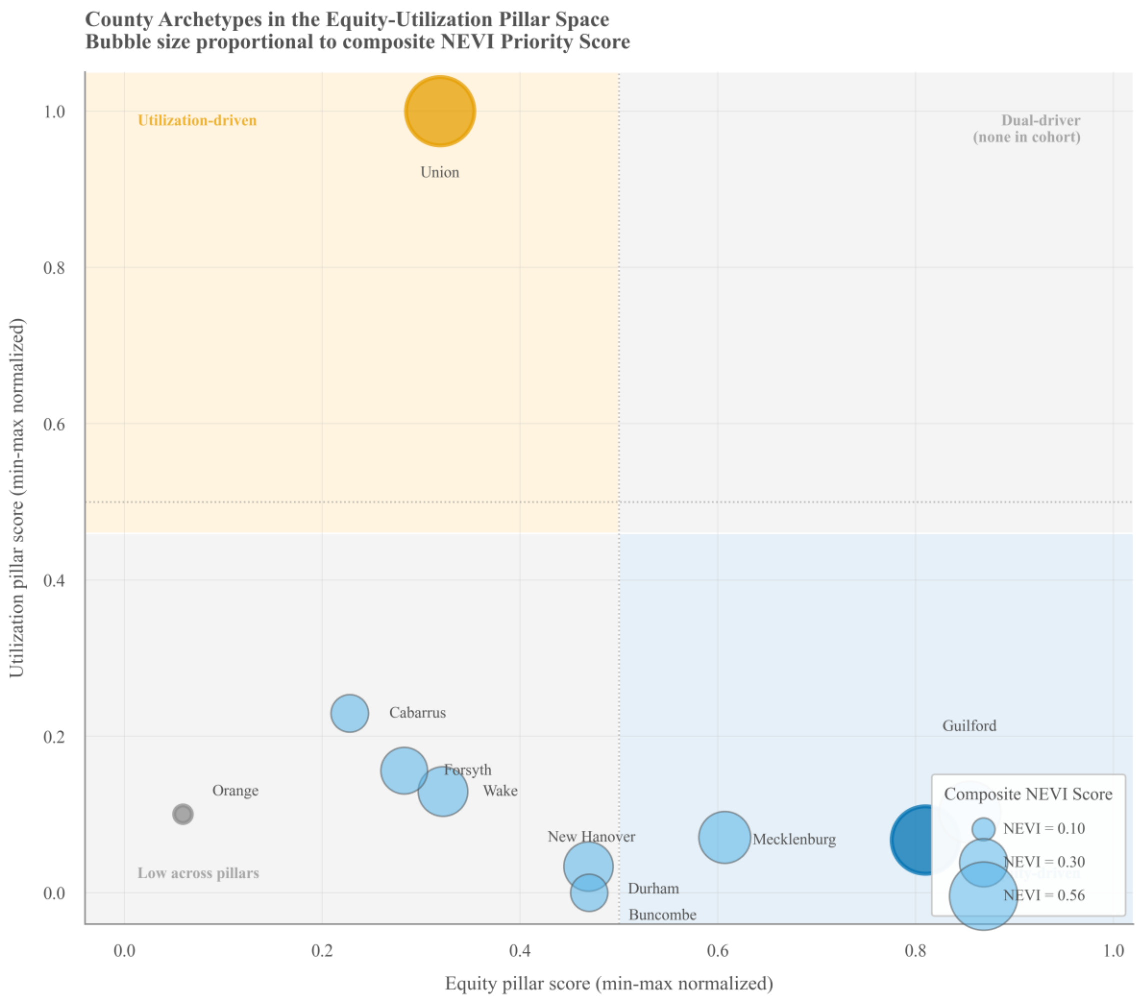

Figure 45

Equity-Utilization Quadrant: Three County Archetypes

Union County (Rank 1, NEVI = 0.561): utilization-driven. Union's 101.5 BEV-per-port ratio is more than three times the next-highest county (Cabarrus at 32.0), producing the maximum normalized utilization score of 1.000 (Section 6.6). This is the signature of a suburban growth corridor where BEV adoption has outpaced infrastructure deployment. Union County is, notably, *not* an employment center: it is a bedroom community of Charlotte with a net commuter outflow of -36,113 per day, meaning much of Union's daytime BEV charging demand actually materializes at Mecklenburg's employment centers. Union's #1 ranking reflects residential under-deployment, so the policy implication is *more stations* rather than *differently targeted stations*.

Mecklenburg County (Rank 2, NEVI = 0.548): equity-driven. Mecklenburg's equity sub-score of 0.810 is the second-highest in the cohort (behind Guilford at 0.855), reflecting concentrated disadvantaged tracts in Charlotte's urban core, paired with the highest cost-effectiveness sub-score (0.801). The latter is driven by the 194,361 net commuters per day arriving at Charlotte's employment centers. Mecklenburg's #2 ranking pairs with the 250-fold within-county density gap (Charlotte Uptown 78.64 ports per 10,000 versus Charlotte East 0.31; Section 6.2) and contributes 41.5% of all within-county inequality in the 10-county cohort (Section 6.3). The policy implication is *better-targeted stations*, not necessarily more stations in aggregate, but specific deployments in the disadvantaged ZIPs that are currently underserved despite Mecklenburg's overall infrastructure depth. Guilford County (Rank 3, NEVI = 0.465) sits in the same archetype with the cohort's highest equity sub-score and moderate utilization strain.

Orange County (Rank 10, NEVI = 0.077): low across all pillars. Orange's Justice40 population-weighted disadvantaged-tract burden is 4.9%, its per-capita charging infrastructure is adequate, and its university-town demographic produces a small EV fleet that existing stations serve well. The policy implication is that Orange does not currently warrant priority NEVI deployment. The reason is not that Orange is uninteresting; it is that federal funding will accomplish more in counties facing utilization pressure or equity gaps. Orange ranks last across all five tested equity weight scenarios (0.30 to 0.50), confirming the position is structurally stable.

The fourth quadrant in Figure 45 (high equity *and* high utilization) is empty in this cohort. No NC county currently combines a high disadvantaged-tract burden with extreme infrastructure under-deployment. The empty quadrant is itself a finding: it identifies the kind of dual-driver county that, if it existed in NC's top-10, would be a clear-cut allocation priority requiring no policy adjudication between competing objectives. That this quadrant is currently empty means every county in the cohort presents a policy trade-off; the scoring framework is designed to make those trade-offs explicit rather than to dissolve them.

7.4 Geographic Sensitivity: Coastal versus Inland

One additional pattern worth noting emerged from the climate sensitivity analysis (Section 4.5). New Hanover County (Rank 4) showed the highest sensitivity to removal of the CEJST climate change category, with its disadvantaged-tract rate dropping by 4.7 percentage points when the category is excluded. Durham (-3.4 percentage points) and Wake (-2.2 percentage points) showed smaller but non-zero sensitivity; the remaining seven study counties showed zero impact. The geographic pattern is consistent with New Hanover's coastal exposure to hurricane and sea-level-rise risk, the FEMA NRI hazard projections that feed CEJST's climate-change category. The implication for policy is narrow but worth surfacing: coastal NC infrastructure-siting decisions are more sensitive than inland decisions to the choice of equity screening methodology. For the broader rankings, the sensitivity is bounded and does not shift the top-3.

7.5 Synthesis

The four discussion subsections converge on a single position. The methodology lanes (§7.1) substantiate the rankings reported in Section 6.6 statistically, geometrically, and via end-to-end traceability. The stakeholder mapping (§7.2) demonstrates that each constituency in the project's value chain receives a defensible artifact appropriate to their decision context. The three-county archetype framing (§7.3, Figure 45) converts the ordered ranking into a policy interpretation that addresses gaps county-level aggregation alone would obscure. The geographic sensitivity analysis (§7.4) bounds the coastal-versus-inland framing without disturbing the top-3. Together, the four subsections supply the prescriptive layer that policymakers can use to inform their own deployment decisions, not to substitute for them.

8. Project Evaluation

This section evaluates the project against the four dimensions specified in the program's project guide: the outcomes delivered, the contribution to the stakeholders the work serves, the contribution to the student's learning, and a self-assessment of the project's execution under sole-author conditions.

8.1 Project Outcomes

This subsection enumerates the deliverable artifacts the project produced and the form in which each is available to a downstream user. The catalog below addresses what a stakeholder receives, not what a stakeholder concludes; the findings these artifacts substantiate are covered in Section 6 and the methodological synthesis that integrates them in Section 7.

Six artifact classes were produced. First, a county-level NEVI Priority Score for the top 10 North Carolina counties by BEV registration count, with rankings stable across the equity-weight sensitivity sweep, the cost-effectiveness sub-weight robustness test, and the cancellation of the remote-work multiplier (Section 4.7.1). Second, a sub-county ZIP-level infrastructure gap analysis usable by local planners for site-selection, with the Theil-T decomposition (Section 6.3) establishing the within-county dominance that justifies the two-tier architecture. Third, a complete analytical pipeline of 27 documented Python scripts, a dependency lockfile (`uv.lock`), and three validation suites (MAPE 4.34% out-of-sample, crosswalk 23 of 23, Phase 4 EDA 59 of 59) that any researcher can re-execute from a clean clone.

The remaining three artifact classes are documentation-focused. Fourth, a Claims-to-Evidence Matrix that ties every numerical assertion in this manuscript to a source file, packaged with the Data Quality Report (Appendix A and `docs/research/data-quality-review-full.pdf`). Fifth, two methodology reviews documenting the internal review processes that substantiated the framework: the five-lane Data Quality Methodology Review (Appendix A) and the four-panelist Manuscript Advisory Panel review (referenced in Appendix B). Sixth, a public GitHub repository (https://github.com/wolfieman/ev-pulse-nc, released May 15, 2026) under the Polyform Noncommercial 1.0.0 license, with 42 publication-grade figures covering all five analytical phases plus the manuscript-prep additions (fig-43 through fig-45).

Each artifact is positioned for use rather than for display. The ranked county list informs allocation rather than dictating it. The ZIP-level gap analysis informs site selection rather than prescribing it. The Python pipeline supports re-execution rather than substituting for verification. The Claims-to-Evidence Matrix substantiates the manuscript rather than replacing reviewer judgment. The collective effect is a decision-support apparatus whose components are independently auditable, an architecture chosen for the federal-program context in which the deliverables are likely to be used.

8.2 Contribution to Stakeholders

Whereas Section 7.2 maps the project's findings to the six stakeholder groups the framework is designed to serve, this section addresses the contribution the project makes to each group's decision context. The project has no corporate client; instead, the stakeholder ecosystem comprises a federal-program administrator (NCDOT and FHWA), federal compliance reviewers (Justice40 auditors and OIG-style assessors), local-jurisdiction planners (county and regional), private-sector operators (charging-network providers), the academic and research community, and the public communities Justice40 is designed to serve. The contributions are tailored to each group's evaluation criteria.

To the NEVI program administrators, the contribution is a county-level priority ranking whose statistical defensibility (VIF 1.41, MAPE 4.34%, triply-robust top-3) is appropriate to the federal-funding decision it is meant to inform. The ranking is a prescriptive layer over the analytical phases, not a replacement for the program-level deliberation that NCDOT's actual deployment process integrates. To Justice40 auditors, the contribution is a 23-of-23 federal-standard crosswalk validation paired with the CEJST v2.0 archival pathway disclosure (Section 5.5) and the climate sensitivity analysis (Section 4.5). The equity pillar's geometry is testable, the data lineage is traceable, and the bounds on the equity claim are stated rather than implied.

To county and regional planners, the contribution is sub-county granularity: the ZIP-level infrastructure gap analysis (Section 6.2) makes the Theil-T decomposition finding (Section 6.3) operationally usable for grant applications and site-selection conversations. To private-sector operators, the contribution is a competitive baseline anchored on the February 2026 AFDC API pull (1,985 stations, 6,145 connectors) and a workplace-charging market sizing from LEHD LODES (859,260 adjusted commuter demand) that supports market-gap analysis at ZIP resolution rather than at the county aggregate.

To the academic and research community, the contribution is the methodological architecture documented in Appendix A and Appendix B: a reproducible analytical pipeline, a Claims-to-Evidence Matrix, the first additive-decomposition application of the Theil index to EV charging infrastructure inequality (Contribution 1 in Section 10.3), and the AI-assisted role-play methodology that operationalized the data quality and manuscript advisory reviews. To Justice40 communities, the contribution is procedural: the allocation methodology is public, the scoring formula is published, the equity pillar carries the largest weight (0.40), and every numerical claim is reproducible from a public GitHub repository. The framework is decision-support across all six groups; the differentiating value to each is the verifiability of the contribution against the group's own evaluation criteria.

8.3 Contribution to Student

This project served as the analytical capstone of my MBA in Business Analytics, and the learning surfaced through it operated at four levels.

At the analytical level, I implemented and verified quantitative methods not covered in prior coursework: the Theil index decomposition (Section 4.4), the Chow structural-break test (Section 4.9), the Variance Inflation Factor diagnostics (Section 4.9), and the area-weighted spatial interpolation between Census-tract and ZCTA geographies (Section 4.5). I researched each method, implemented it in Python, validated it against published benchmarks, and integrated it into the scoring framework. The Theil decomposition in particular required moving beyond the textbook treatment of inequality measurement (where the Gini coefficient typically suffices) into the literature on additively decomposable measures (Bourguignon, 1979; Shorrocks, 1980; Theil, 1967) and the small but growing applied literature in transportation research (Choi, Xu, & Jiao, 2025).

At the methodological level, I designed internal review processes appropriate to a sole-author research project. The five-lane data quality methodology review and the four-panelist manuscript advisory panel (Appendix B) operationalized AI-assisted role-play as a structured review mechanism, with each panel anchored to a fixed repository commit and to a pre-registered claims matrix. I designed the methodology during the project, refined it across iterations, and documented it as a contribution in its own right.

At the federal-program level, I learned the structure and current state of the NEVI Formula Program (FHWA Final Rule, February 2023), the Justice40 Initiative (Executive Order 14008 and the subsequent CEJST tooling and methodology), the CEJST v2.0 design and its January 22, 2025 decommissioning, and the LEHD LODES commuter data ecosystem maintained by the U.S. Census Bureau. Federal-program literacy is rarely taught in MBA coursework; I had to acquire it against the actual policy artifacts as they evolved during the analytical window.

At the project-management level, I executed a five-phase analytical architecture as a sole researcher across an academic semester, with each phase resolving into committed code, processed data, validation evidence, and figure outputs that the next phase could build on without rework. Three project-management disciplines mattered most. The first was phase-gating: treating each phase boundary as a commit checkpoint that the next phase could assume rather than verify. The second was scope discipline: validating each phase's outputs against a pre-registered claims matrix before consuming them downstream, and resisting the temptation to revise prior-phase work when new findings emerged. The third was explicit dependency management: every artifact a downstream phase needed was documented as a phase deliverable rather than a side effect of running prior scripts.

8.4 Self-Assessment

This section assesses the execution of the project from the sole-author perspective the work was conducted under. The project's principal execution strengths and the principal execution constraints are identified explicitly so that any reader can weigh the work product against the conditions in which it was produced.

The project's execution strengths center on three architectural choices. The first is the five-phase analytical pipeline (Phase 1 forecast validation through Phase 5 equity overlay), which scoped each analytical commitment to a single tractable component with a defined output, allowing the next phase to assume the prior phase's artifacts as input rather than re-derive them. The second is the AI-assisted role-play methodology documented in Appendix B, which operationalized multi-perspective review (data quality, federal compliance, methodology, business strategy) for a sole researcher in a way that internal solo review could not have matched. The third is the direction-of-bias disclosure pattern adopted in the Limitations section (Section 9), in which every named limitation states the direction of its potential effect on the headline findings rather than merely acknowledging the limitation's existence.

The project's execution constraints surface as honestly. The Theil novelty claim required narrowing after a literature check identified Choi, Xu, and Jiao (2025), and the manuscript's contribution claim was tightened accordingly (Contribution 1 in Section 10.3, with the underlying lit-check preserved at `docs/research/prior-work-theil-ev-charging.md`). The 10-county scope is a constraint imposed by the analytical window rather than a design choice; the framework is engineered for statewide extension, and the statewide application is the first item in the Future Work section (Section 11.1). The internal methodology reviews are not external peer review, and the disclosure of that boundary in Appendix B is the honest framing of an apparatus that was chosen for its operational tractability rather than because it substitutes for external assessment.

9. Limitations

This section enumerates the principal limitations of the study. The framing principle is the one Expert 3 of the data quality panel articulated cleanly during the methodology review: *a limitation with a direction is a disclosure; a limitation without one is a wound.* Every limitation below names a direction of bias: works against the headline finding, works toward it, cancels mathematically, or is bounded with no directional effect. Each direction is paired with a future-work pointer. Some limitations reinforce the headline findings (Section 6) rather than undermine them; others bound the analytical scope; a third group identifies federal-policy risks that quantitative reviewers will not flag but federal-program reviewers will. The ten limitations below are grouped by direction of bias: limitations that reinforce the headline findings (§§9.1–9.4), limitations bounded with no directional bias on the rankings or undetermined pending additional data (§§9.5–9.7), and federal-policy or program risks (§§9.8–9.10). They emerged from the data quality panel (Round 1, April 2026) and the manuscript advisory panel (May 2026).

9.1 Top-10 County Selection Limits Generalizability

The scoring framework is computed for the top 10 NC counties by BEV registration count rather than all 100 counties. The 10-county cohort captures 73% of the statewide BEV fleet; the framework is engineered for statewide extensibility but is currently validated only within this cohort. The decision was substantiated as intentional by the data quality panel: Expert 1 confirmed the scope (73% BEV coverage is the natural majority of the decision space), and Expert 4 confirmed the within-cohort statistical stability (triply-robust top-3, Section 4.7.1). The 10-county selection over-represents urban and suburban infrastructure dynamics relative to a statewide allocation; this is countered by the Theil-T decomposition (Section 6.3), which shows that 84.5% of charging infrastructure inequality lives within counties, the dynamic the 10-county scoring explicitly targets. *Direction of bias: bounded by the Theil finding and the weight sensitivity analysis. Future work:* statewide extension to all 100 counties, which would activate the two currently-dormant sub-metrics (Section 9.5) and provide cross-validation against the within-cohort top-3 ranking.

9.2 LODES 2021 Vintage Predates Full Return-to-Office

The workplace charging demand model (Section 4.6) uses LEHD LODES 2021 origin-destination commuter flows, which captures commuting patterns from a period when COVID-era remote work was still elevated relative to pre-pandemic norms. LODES 2021 is the most recent public release; CTPP 2016–2020 would be strictly older; commercial alternatives (StreetLight, Replica) are $10,000 to $100,000 per state and out of scope. The 0.85 remote-work multiplier (Barrero, Bloom, & Davis, 2023) handles the average effect at a national level. *Direction of bias: cancels mathematically in min-max normalization.* Expert 4 of the data quality panel demonstrated empirically that rankings are identical at 0.75, 0.85, and 0.95 multiplier values (Section 4.7.1). The vintage affects absolute demand estimates (for example, the 859,260 statewide adjusted demand figure in Section 6.4) but not the county ordering that drives the NEVI recommendation. *Future work:* county-specific remote-work rates when ACS post-COVID estimates become reliable at acceptable margins of error.

9.3 CEJST One-Strike Design May Over-Identify Disadvantaged Tracts

CEJST v2.0 uses a one-strike design: a Census tract is classified as disadvantaged if it exceeds the 90th percentile on any single indicator across eight burden categories, provided the low-income threshold is also met. This intentionally casts a wide net (Section 3.4). A reviewer could legitimately ask whether the resulting 43.0% NC disadvantaged-tract rate is inflated by the one-strike rule, biasing the equity pillar in favor of equity-driven counties (Mecklenburg, Guilford). The climate sensitivity analysis in Section 4.5 addresses this directly: removing the climate change category entirely (the only category that incorporates forward-looking model projections) drops the statewide rate from 43.0% to 37.3%, essentially matching the national average. Seven of ten study counties show zero impact; the three affected counties (New Hanover, Durham, Wake) shift by 2.2 to 4.7 percentage points. *Direction of bias: bounded; reinforces robustness rather than undermining the equity rankings.* The equity rankings are driven by empirical health, housing, and workforce indicators, not by the climate projections. *Future work:* re-examine against any federal successor screening instrument when available.

9.4 Static BEV Registration; No Growth Trajectory Modeling

The framework treats BEV registrations at their October 2025 levels rather than modeling continuing growth trajectories through the NEVI deployment horizon. Phase 1 validation (Section 6.1) quantifies forecast accuracy at MAPE 4.34% on a four-month out-of-sample holdout, and the 69.00% underprediction rate is explained by the August 2022 IRA structural break (Chow F = 1,268.35; Section 4.9). The forecast accuracy is sufficient for county-level capital allocation as of the analysis window, but does not project continuing acceleration. *Direction of bias: under-states demand growth*, which means the framework's BEV-per-port utilization scores are conservative; the actual port shortfall in high-growth counties will be larger by the time NEVI-funded stations are operational. This direction reinforces the urgency in top-ranked counties rather than overstating it. *Future work:* longitudinal growth-trajectory modeling using post-IRA training windows; ensemble forecast combinations across the ESM, ARIMA, and UCM model families.

9.5 Two Zero-Variance Sub-Metrics at 10-County Scope

At the 10-county scope, two of the nine designed sub-metrics carry zero variance: `zero_station_pct` is 0.0 for all 10 counties (all top-10 urban counties have charging infrastructure in every inhabited ZIP), and `forecast_buffer` is a uniform 0.045 applied globally (Section 4.7). Both sub-metrics would contribute differentiation at statewide scale, where rural counties have zero-station ZIPs and county-specific forecast errors could replace the global buffer. Seven of the nine designed sub-metrics actively differentiate counties at the 10-county scope, which is sufficient for the within-cohort ranking. *Direction of bias: no directional effect; bounded scope.* The dormant sub-metrics do not bias the rankings in any direction; they simply do not contribute information for this cohort. *Future work:* statewide extension activates both sub-metrics.

9.6 No Charging Behavior Data Beyond AFDC

The Alternative Fuels Data Center provides station-level infrastructure location, charging level, network, and access type, not real-time utilization data (session frequency, duration, energy delivered). The BEV-to-port ratio used as the primary demand-supply metric is therefore a proxy for utilization pressure, not a measured throughput. Private-sector operator data (ChargePoint, Tesla) would refine this but is not public; the DOE EVSE data ecosystem gap on session-level telemetry is a known limitation of the broader research landscape, not specific to this study. *Direction of bias: undetermined.* Without observed utilization, high BEV-per-port ratios may reflect either genuine under-deployment (the assumed direction) or low utilization despite ports being available. The Cary, NC utilization study (Kutela et al., 2024) provides limited counter-evidence that NC stations are heavily used in the locations studied, but the data does not generalize to the top-10 county scope. *Future work:* acquire or simulate session-level utilization data; integrate proxy throughput estimates from network operators where available.

9.7 Area-Weighted Interpolation Assumes Uniform Within-Tract Population

The tract-to-ZCTA crosswalk (Section 4.5) uses area-weighted interpolation, which assumes uniform population distribution within tracts. This assumption holds reasonably well in urban and suburban tracts where tracts are small and land use is dense; the assumption is weaker in large rural tracts that span heterogeneous land uses. EPA EJScreen and HUD USPS Crosswalk both use area-weighted interpolation as their federal standard, so this study is methodologically aligned with federal practice. The crosswalk validation suite (Section 4.5; 23 of 23 checks passed) confirms per-ZCTA area conservation within 1% of the original tract area. *Direction of bias: bounded; the 10-county study area is urban and suburban, the population regime where area-weighted interpolation is most accurate. The bias is likely small for the rankings reported in Section 6. Future work:* dasymetric (population-weighted) interpolation using gridded population data; spatial sensitivity analysis comparing area-weighted versus dasymetric outputs for the equity pillar.

9.8 CEJST Methodology Lock at v2.0

The Justice40 disadvantaged-tract designations used in the equity pillar are anchored on CEJST v2.0 (December 2024 release) as it stood on January 21, 2025, the last operational day before the federal screening tool was removed from `screeningtool.geoplatform.gov` following the rescission of Executive Order 14008 on January 22, 2025. The data continuity is preserved through the EDGI / Public Environmental Data Partners community archive (Section 5.5), and the methodological continuity is preserved through the CEJST v2.0 Technical Support Document published by the Council on Environmental Quality, which remains the federal record of the screening criteria. The methodology version itself is locked at v2.0; any future federal successor (whether a restored CEJST, an EJScreen-based replacement, or a state-level instrument) may produce different disadvantaged-tract classifications. *Direction of bias: unknown, depends on successor methodology.* The methodology document anchors the analytical operation; the data is preserved via the community archive with chain-of-custody documentation. Whether a successor instrument produces higher, lower, or differently-distributed disadvantaged-tract rates is not currently determinable. *Future work:* re-run the equity-pillar analysis against any federal successor instrument when available, with the understanding that the underlying disadvantaged-community designations may shift in either direction.

9.9 NEVI Program Rule Evolution Risk

The framework is anchored on the NEVI program rule as of the analysis date (FHWA Final Rule, February 2023, as referenced in Sections 3.4 and 4.5). If FHWA revises the rule on items such as minimum station spacing along Alternative Fuel Corridors, minimum port count per station, technology requirements (the 150 kW DCFC threshold), or eligibility criteria for community charging, the cost-effectiveness pillar inputs and the deployment-readiness assumptions shift. This is a federal-program-specific risk that quantitative reviewers will not typically flag but federal-program reviewers and NEVI administrators will. *Direction of bias: unknown, rule-dependent.* A loosened rule (lower minimum port count, slower deployment expectations) would reduce the cost-effectiveness pillar's discrimination across counties; a tightened rule would increase it. Neither direction is currently known. *Future work:* monitor FHWA rulemaking and rerun the framework when material rule changes occur; consider regime-indicator variables in any future statewide extension.

9.10 State-Administered Formula Program Variance

NEVI is federally funded but state-administered. NCDOT's actual deployment decisions may diverge from this study's scoring output for legitimate non-equity reasons: right-of-way constraints, utility-coordination timelines, station-host availability, supply-chain timing for electrical equipment, and political prioritization at the state and county levels. The framework is decision-support, not decision-making. NCDOT's February 2026 shift from a corridor-only NEVI strategy toward 16 rural and community locations (cited in Section 1.2) is the kind of policy decision the scoring framework does not predict and is not designed to substitute for. The framework's value is to inform NCDOT's prioritization process, not to replace it. *Direction of bias: no statistical bias; framing risk only.* This is the most important sentence in the limitations section: it pre-empts the "your scoring didn't match actual deployments" critique that would follow any Phase 7 NCDOT deployment validation (Section 11). A divergence between this study's rankings and NCDOT's actual deployments does not falsify the framework; it identifies the legitimate non-equity factors that NCDOT's decision process integrates and that this study explicitly does not. *Future work:* Phase 7 NCDOT deployment validation; case-study analysis of divergences when deployment data becomes available.

9.11 Synthesis

Across the ten limitations enumerated above (§§9.1–9.10), four reinforce the headline findings (§§9.1–9.4: bounded, cancels mathematically, bounded with robustness evidence, and conservative direction respectively); two are bounded with no directional bias on the rankings (§§9.5, 9.7); one (§9.6) is undetermined pending session-level utilization data; and three are federal-policy or program risks whose impact is rule- or successor-dependent (§§9.8–9.10). No limitation in this list directly undermines the headline rankings reported in Section 6. The systematic application of the direction-of-bias disclosure pattern is what converts this section from a defensive list into a credibility-positive one: every limitation is named, every direction is stated, every future-work pointer is identified. A reviewer who reads this section should walk away with a clearer understanding of where the framework's claims hold up, where they are bounded, and where they depend on federal-policy stability that the study cannot guarantee.

10. Conclusion

10.1 Findings in Brief

This study delivered a defensible county-level NEVI Priority Score for the top 10 North Carolina counties by BEV registration count, with Union (0.561), Mecklenburg (0.548), and Guilford (0.465) leading the ranking and the top-3 set stable across all five tested equity-weight scenarios. Out-of-sample forecast validation produced a mean absolute percentage error of 4.34% across 400 county-month observations, and a Chow structural-break test identified the August 2022 Inflation Reduction Act passage as a strongly significant break ($F = 1{,}268.35$, $p < 1 \times 10^{-6}$), reframing the 69.00% systematic underprediction as the statistical fingerprint of accelerated post-IRA adoption rather than a model failure. The scoring framework's three pillars are statistically independent (maximum Variance Inflation Factor of 1.41 against a conventional 5.0 concern threshold), and a Theil-T decomposition revealed that 84.5% of charging infrastructure inequality across the top-10 counties is *within* counties rather than between them, a finding that motivated the two-tier scoring architecture (county-level ranking plus ZIP-level targeting) and that has not, to the author's knowledge, been previously applied as a structural decomposition to EV charging infrastructure. A spatial overlay of charging stations on CEJST-designated Justice40 disadvantaged tracts (CEJST v2.0, as it stood on January 21, 2025, the last operational day before the federal screening tool was decommissioned) found that 24.5% of stations within the top-10 county scope sit in disadvantaged tracts, approximately proportional to the 18.5% population-weighted disadvantaged-tract share within those counties.

10.2 Summary of Findings

The findings from each of the five analytical phases (defined in Section 4.1) are summarized below, with the NEVI scoring framework integrating them into a single prescriptive output. **Phase 1** validated the underlying SAS Model Studio BEV forecasts against a four-month out-of-sample holdout (July through October 2025), confirming usability for capital allocation decisions and identifying the IRA structural break as the substantive explanation for systematic underprediction. **Phase 2** upgraded the supply-side infrastructure baseline from a DCFC-only July 2024 extract (355 stations, 1,725 connectors) to a full February 2026 AFDC API pull capturing all charging levels and access types (1,985 stations, 6,145 connectors across 358 ZIP codes and 267 cities). **Phase 3** measured ZIP-level infrastructure inequality at both Gini-scalar (statewide population-weighted 0.566) and Theil-T-decomposition levels (84.5% within-county, 15.5% between-county), establishing the empirical justification for the two-tier scoring architecture. **Phase 4** layered workplace charging demand onto residential registration data using LEHD LODES 2021 origin-destination commuter flows filtered through a three-layer adjustment pipeline (SE03 income filter, ACS household-income correction at the $75,000 threshold, and a 0.85 remote-work multiplier), producing an adjusted statewide workplace demand estimate of 859,260 workers. **Phase 5** integrated CEJST v2.0 Justice40 designations into the equity pillar via an area-weighted tract-to-ZCTA crosswalk aligned with EPA EJScreen and HUD USPS Crosswalk federal-standard methodology, with a climate sensitivity analysis demonstrating that 7 of 10 study counties are unaffected by removing the most-contested CEJST category. The **NEVI scoring framework** combined the validated demand-side forecasts, the upgraded supply-side baseline, the within-county inequality finding, the workplace-charging demand layer, and the Justice40 equity overlay into a composite Priority Score (weights 0.40 Equity / 0.35 Utilization / 0.25 Cost-Effectiveness) ranking the top 10 NC counties for NEVI Formula Program funding consideration. A four-panelist manuscript advisory panel and a separate five-panelist data quality methodology review (60-plus items reviewed; zero open at the time of submission) substantiated the framework's defensibility across data engineering, spatial geometry, federal compliance, econometric, and business-strategy lenses.

10.3 Five Unique Contributions

This study makes five contributions to the EV infrastructure allocation literature, each substantiated by a specific lane of the methodology review.

Contribution 1: First additive-decomposition application of the Theil index to EV charging infrastructure inequality, integrated into an allocation framework. Prior work using the Theil index in this domain (notably Choi, Xu, & Jiao, 2025, in *Transportation Research Part D*) employs Theil as one scalar inequality metric alongside the Lorenz curve, Gini coefficient, and Palma ratio. The methodological contribution here is the exploitation of the **additive decomposability of the GE(1) class** (Bourguignon, 1979; Shorrocks, 1980) to partition EV charging infrastructure inequality into its between-county and within-county components, paired with the use of those components as a design input to the prioritization framework (Section 3.3, Section 4.4). The contribution is not "first Theil applied to EV charging" but "first **structural decomposition** of Theil applied to EV charging *and integrated with* a county-level allocation architecture." Substantiated by Expert 4 (statistical robustness of the decomposition: machine-precision verification, 84.5% within-county confirmed by a Theil-L robustness check at 82.5%) and Expert 2 (spatial hygiene of the underlying ZCTA infrastructure data).

Contribution 2: First application of the HUD USPS / area-weighted crosswalk methodology to integrate CEJST Justice40 designations with AFDC station data at ZCTA resolution in North Carolina. Area-weighted spatial interpolation between Census-tract and ZCTA geographies is a HUD/EPA federal standard, not a novel method. The contribution claimed here is the **integration target**: applying that standard methodology to overlay CEJST v2.0 disadvantaged-community designations onto AFDC charging-station inventory at the sub-county resolution required for state-level NEVI prioritization, with a 12-check three-tier validation suite (23 of 23 sub-checks passed; Section 4.5) backstopping the geometric integrity. Substantiated by Expert 1 (data engineering: join-rate integrity, crosswalk loss validation) and Expert 2 (spatial: CRS discipline across the EPSG:32119 / EPSG:4326 / EPSG:4269 stack, per-ZCTA area conservation within 1%).

Contribution 3: NEVI-specific three-pillar scoring formulation with documented VIF orthogonality and triply-robust top-3 stability under weight perturbation. Multi-criteria scoring frameworks are well established in transportation infrastructure prioritization (Section 3.6); USDOT RAISE and BIL CFI are direct federal precedents. The contribution here is not the multi-criteria structure but the **validation properties** of the specific NEVI formulation: a Variance Inflation Factor maximum of 1.41 across the three pillars confirms statistical independence (Section 4.9); the top-3 ranking (Union, Mecklenburg, Guilford) is stable across the equity-weight sensitivity sweep, the cost-effectiveness sub-weight robustness test, and the mathematically demonstrated cancellation of the remote-work multiplier (the "triply-robust" framing, Section 4.7.1). Substantiated by Expert 4 (VIF analysis, Chow test, robustness testing) and Expert 5 (the three-archetype interpretation that converts the ranking into a policy conversation, Section 7.3).

Contribution 4: Empirical validation that within-county inequality dominates between-county inequality in NC charging infrastructure (84.5% Theil). The within-county dominance finding is, in the broader transportation-equity literature, the kind of structural diagnostic that allocation policy often assumes but rarely measures directly for a specific federal-program-relevant infrastructure type. The 84.5% within-county share, verified to machine precision through the additive Theil-T decomposition and confirmed by a Theil-L (GE(0)) robustness check at 82.5%, supplies the empirical justification for the two-tier architecture that distinguishes this framework from county-only allocation formulas (Sections 3.3, 4.4, 6.3). Substantiated by Expert 2 (spatial input integrity to the decomposition) and Expert 4 (statistical robustness of the decomposition itself).

Contribution 5: Climate sensitivity analysis demonstrating CEJST robustness for the NC context. CEJST v2.0's eight burden categories include exactly one that incorporates forward-looking model projections: climate change (FEMA NRI hazard estimates and First Street Foundation flood projections). The sensitivity analysis re-ran the equity-pillar computation with the climate change category removed entirely and found that 7 of 10 study counties show zero impact, while the three affected counties (New Hanover, Durham, Wake) shift by 2.2 to 4.7 percentage points, within bounds that do not alter the top-3 ranking. The contribution is specific: a documented demonstration that the equity rankings produced under CEJST v2.0 are not artifacts of the most-contested CEJST design choice (Section 4.5). Substantiated by Expert 3 (federal compliance: CEJST archival pathway and CEQ Technical Support Document anchoring) and Expert 2 (sensitivity execution).

10.4 Closing Synthesis

Federal infrastructure programs increasingly require multi-criteria prioritization frameworks that reconcile efficiency with equity, and that do so in a way reviewers in both the academic and federal-program communities can verify. The NEVI Priority Score formulation presented here supplies a prescriptive layer that makes findings actionable for policymakers. It rests on a validated forecast layer (MAPE 4.34%) and a structurally robust scoring matrix (VIF 1.41 maximum, with triply-robust top-3 stability). The equity overlay is geometrically sound, passing 23 of 23 federal-standard crosswalk checks, and the additive-decomposition diagnostic supplies the empirical finding that 84.5% of charging-infrastructure inequality is within-county rather than between-county. The framework is decision-support, not decision-making: NCDOT's actual NEVI deployment decisions will and should incorporate factors this study does not model (right-of-way, utility coordination, supply-chain timing, political prioritization). What the framework supplies is the analytical floor a state's NEVI allocation can stand on: a floor that is, to the best of this work's verification capacity, both data-driven and policy-aligned, and that can be reproduced from a public GitHub repository by anyone willing to re-execute the published scripts against the committed data.

11. Future Work

The limitations documented in Section 9 each carry an associated future-work pointer. This section consolidates those pointers and adds research directions that are out of scope for the current analytical window but would meaningfully extend the framework. The items are organized into five sections: geographic and data-scope extensions, validation and verification studies, methodological refinements, federal-policy responsiveness, and a closing synthesis.

11.1 Scope and Data Extensions

The most immediate extension is **statewide application to all 100 NC counties**. The framework is engineered for this. Every data pipeline (NCDOT BEV registrations, AFDC infrastructure, LEHD LODES commuter flows, CEJST tract designations, ACS demographics) processes all 100 counties; only the scoring layer is scoped to the top 10 (Section 9.1). A statewide run would activate the two currently-dormant sub-metrics (`zero_station_pct` and a per-county `forecast_buffer` replacing the global 0.045; Section 9.5), increase the effective scoring resolution from seven to nine of nine sub-metrics, and provide cross-validation against the within-cohort top-3 ranking reported in Section 6.6.

Two data-acquisition extensions would further refine the analytical resolution. **ZIP-code-level BEV registration data**, if NCDOT publishes it or if the project obtains licensed commercial data (IHS Markit, Experian VIO; estimated $5,000–$20,000 per state), would enable BEV-to-port ratios at the ZIP level rather than only at the county level, removing the two-scale mismatch documented in Section 5.6. **Interstate commuter inclusion via LODES auxiliary files** would close the 4.6% gap between the LODES OD Main file (intra-state only) and the WAC file (workplace counts including cross-border commuters; Section 5.4), particularly relevant for Mecklenburg (Charlotte–South Carolina corridor) and the Virginia- and Tennessee-line border counties.

11.2 Validation and Verification

Phase 7 NCDOT deployment validation is the most consequential external check the framework can receive. NCDOT's February 2026 shift toward 16 rural and community NEVI locations (Section 1.2) signals that actual deployment data will become available after the planned RFP cycle. Comparing this study's scoring output against NCDOT's actual deployment decisions would test whether the equity-weighted methodology captures real policy priorities; agreement would substantiate the framework, and disagreement would identify the legitimate non-equity factors NCDOT's decision process integrates that this study does not model (Section 9.10). Either outcome is informative.

Extended validation window with formal break testing. The Phase 1 validation rests on a four-month out-of-sample holdout (July through October 2025; n = 400 county-month observations). Extending the holdout to twelve months (July 2025 through June 2026) would substantially increase statistical power and enable a formal Chow test of the May 2025 NCDOT methodology change, the test that is currently degenerate at the 2025 break date ($n_2 = k = 2$; Section 4.9) because too few post-break observations existed at the analysis window.

Composite-score confidence intervals. The framework currently reports pillar-level weight sensitivity (Section 4.7.1, the "triply-robust" finding) but does not propagate the underlying uncertainty sources (forecast prediction-interval widths, LEHD LODES noise infusion, Gini sampling variability, and CEJST classification thresholds) into a single composite NEVI score confidence interval. A Monte Carlo simulation or analytical error-propagation pass would produce results of the form "Union ranks #1 with 90% probability under the propagated uncertainty," which would strengthen the framework's decision-support value for stakeholders comparing closely-ranked counties.

11.3 Methodological Refinements

Longitudinal growth-trajectory modeling would address the static BEV-registration treatment documented in Section 9.4. Rather than treating BEV counts at their October 2025 levels, a forward-projection model calibrated on the post-IRA growth regime identified by the August 2022 Chow break would generate county-level trajectories through the NEVI deployment horizon. The current framework's conservative direction (the BEV-per-port utilization scores understate the actual port shortfall by the time NEVI-funded stations are operational) becomes quantifiable rather than directional.

Dasymetric (population-weighted) tract-to-ZCTA interpolation would replace the area-weighted interpolation currently used in the equity pillar (Section 4.5). Area-weighted is the EPA EJScreen / HUD USPS Crosswalk federal standard, and the per-ZCTA area conservation is within 1% (Section 4.9), but a population-weighted refinement using a gridded population product (LandScan, WorldPop) would tighten precision in large rural tracts that span heterogeneous land uses (Section 9.7). For the urban-and-suburban top-10 cohort, the refinement is modest; for a statewide extension, it would matter more.

County-specific remote-work rates would replace the uniform 0.85 multiplier in the workplace-charging demand pipeline (Section 4.6). The current multiplier cancels mathematically in min-max normalization (Section 9.2), meaning rankings are invariant. But absolute demand estimates and per-county capacity-planning recommendations depend on the multiplier value. When ACS post-COVID estimates of county-level remote-work rates become available with acceptable margins of error, replacing the national average with per-county rates would refine the cost-effectiveness pillar's absolute scale.

Full design-of-experiments coverage of the weight simplex. The current weight sensitivity analysis is one-factor-at-a-time (OFAT; Section 4.7.1): equity weight varies across 0.30–0.50 while the utilization-to-cost-effectiveness ratio is held constant. A formal fractional factorial or response-surface design over the entire weight simplex (the constrained 2D triangular region where the three weights sum to 1.0) would identify whether the current 0.40/0.35/0.25 weights sit near a discrimination optimum or on a steep gradient. The triply-robust top-3 finding under OFAT suggests the gradient is gentle, but a full DOE pass would confirm.

11.4 Federal-Policy Responsiveness

CEJST successor re-analysis. The equity-pillar findings are anchored on CEJST v2.0 as of January 21, 2025 (Sections 5.5, 9.8). Any future federal successor (a restored CEJST, an EJScreen-based replacement, or a state-level instrument) may produce different disadvantaged-tract classifications. Re-running the equity-pillar analysis against any successor instrument when one becomes available, with the methodology document (CEQ Technical Support Document) as the anchor for tracking changes, would test the durability of the equity rankings across federal-policy regimes.

NEVI program rule monitoring. The framework is anchored on the FHWA NEVI Final Rule as of the analysis date (Section 9.9). Material rule changes (to minimum port count, minimum station spacing, technology requirements, eligibility criteria) would shift the cost-effectiveness pillar's inputs and potentially the rankings. Re-running the framework when FHWA issues a revised rule is a maintenance task; tracking the rulemaking process is a research-monitoring task that does not require new analysis but does require attention.

11.5 Synthesis

The future-work items above are ordered from immediate-extension (statewide application; Phase 7 NCDOT validation) to longer-horizon refinements (dasymetric interpolation, full DOE simplex coverage) to federal-policy-responsive maintenance (CEJST successor, NEVI rule monitoring). None of these extensions is required for the headline findings reported in Section 6 to stand; each represents a direction in which the framework can be strengthened, broadened, or kept current with federal-policy evolution. The framework's design discipline (Section pointers for every cross-reference, the Claims-to-Evidence Matrix in Appendix A, the public GitHub repository) is intended to make these extensions tractable for any researcher willing to take them up.

12. References

Cited Works

Barrero, J. M., Bloom, N., & Davis, S. J. (2023). The evolution of work from home. *Journal of Economic Perspectives*, 37(4), 23–49. https://doi.org/10.1257/jep.37.4.23

Bourguignon, F. (1979). Decomposable income inequality measures. *Econometrica*, 47(4), 901–920. https://doi.org/10.2307/1914138

Celik, S., & Ok, S. (2024). Electric vehicle charging stations: Model, algorithm, simulation, location, and capacity planning. *Heliyon*, 10(7), e29153. https://doi.org/10.1016/j.heliyon.2024.e29153

Chang, D., Erstad, D., Lin, E., Rice, A., & Tsao, A. (2012). *Financial viability of non-residential electric vehicle charging stations.* UCLA Luskin Center for Innovation. https://innovation.luskin.ucla.edu/wp-content/uploads/2019/03/Financial_Viability_of_Non-Residential_EV_Charging_Stations.pdf

Choi, S. J., Xu, Y., & Jiao, J. (2025). Utility or equity? Analyzing public electric vehicle charger allocations in Austin, Texas. *Transportation Research Part D: Transport and Environment.* https://www.sciencedirect.com/science/article/abs/pii/S1361920925004043

Council on Environmental Quality. (2024). *Climate and Economic Justice Screening Tool (CEJST) version 2.0: Technical Support Document.* Executive Office of the President of the United States. (Archived methodology document; the operational tool at `screeningtool.geoplatform.gov` was decommissioned January 22, 2025 following the rescission of Executive Order 14008.)

Hakimi, S. L. (1964). Optimum locations of switching centers and the absolute centers and medians of a graph. *Operations Research*, 12(3), 450–459. https://doi.org/10.1287/opre.12.3.450

International Energy Agency. (2024). *Global EV Outlook 2024: Moving towards increased affordability.* IEA Publications. https://www.iea.org/reports/global-ev-outlook-2024

Joint Office of Energy and Transportation. (2024). *State EV registration and infrastructure data.* U.S. Department of Energy and U.S. Department of Transportation. https://driveelectric.gov/states

Kutela, B., Shita, H., Salum, J. H., Mlawa, D., Das, A., & Chimba, D. (2024). Exploring the utilization of electric-vehicles' free public charging facilities in small towns: A case of the town of Cary, North Carolina. *Case Studies on Transport Policy*, 15, 101142. https://doi.org/10.1016/j.cstp.2023.101142

Nilsson, G., Owen Aquino, A. D., Coogan, S., & Molzahn, D. K. (2024). GreenEVT: Greensboro Electric Vehicle Testbed. *IEEE Systems Journal*, 18(1), 584–595. https://doi.org/10.1109/JSYST.2023.3343692

North Carolina Department of Transportation. (2025). *Monthly vehicle registration counts by county: Battery electric vehicles, September 2018–October 2025.* Climate Change Initiatives. https://www.ncdot.gov/initiatives-policies/environmental/climate-change/Pages/default.aspx

North Carolina Department of Transportation. (2026, March 27). *North Carolina received $109 million in NEVI funding to build out EV infrastructure along a network of approved interstates and highway corridors and in communities* [Press release].

https://www.ncdot.gov/news/press-releases/Pages/2026/2026-03-27-applications-open-second-wave-ev-charging-stations.aspx

ReVelle, C. S., & Swain, R. W. (1970). Central facilities location. *Geographical Analysis*, 2(1), 30–42. https://doi.org/10.1111/j.1538-4632.1970.tb00142.x

Shorrocks, A. F. (1980). The class of additively decomposable inequality measures. *Econometrica*, 48(3), 613–625. https://doi.org/10.2307/1913126

Theil, H. (1967). *Economics and information theory.* North-Holland Publishing.

Uimonen, S., & Lehtonen, M. (2020). Simulation of electric vehicle charging stations load profiles in office buildings based on occupancy data. *Energies*, 13(21), 5700. https://doi.org/10.3390/en13215700

U.S. Census Bureau, Center for Economic Studies. (2021). *LEHD Origin-Destination Employment Statistics (LODES), version 8.1: North Carolina origin-destination main file.* Longitudinal Employer-Household Dynamics Program. https://lehd.ces.census.gov/data/

U.S. Congress. (2021). *Infrastructure Investment and Jobs Act of 2021 (IIJA), Pub. L. 117–58, §11401: National Electric Vehicle Infrastructure Formula Program.* Codified at 23 U.S.C. §175.

U.S. Department of Energy. (2024). *Alternative Fuels Data Center: Electric vehicle registration and charging infrastructure statistics.* https://afdc.energy.gov/

Wood, E., Rames, C., Muratori, M., Srinivasa Raghavan, S., & Melaina, M. (2017). *National plug-in electric vehicle infrastructure analysis* (NREL/TP-5400-69031). National Renewable Energy Laboratory. https://docs.nrel.gov/docs/fy18osti/70779.pdf

Zafar, U., Bayram, I. S., & Bayhan, S. (2021). A GIS-based optimal facility location framework for fast electric vehicle charging stations. In *2021 IEEE 30th International Symposium on Industrial Electronics (ISIE)* (pp. 1–6). https://doi.org/10.1109/ISIE45552.2021.9576448

Additional Sources Consulted

The following works informed the analytical approach, methodology, or contextual understanding of this study but are not directly cited in the body above. They are included to document the breadth of the literature review.

Allison, P. D. (1978). Measures of inequality. *American Sociological Review*, 43(6), 865–880. https://doi.org/10.2307/2094626

Canepa, K., Hardman, S., & Tal, G. (2019). An early look at plug-in electric vehicle adoption in disadvantaged communities in California. *Transport Policy*, 78, 19–30. https://doi.org/10.1016/j.tranpol.2019.03.009

Ceriani, L., & Verme, P. (2012). The origins of the Gini index: Extracts from *Variabilità e Mutabilità* (C. Gini, 1912). *Journal of Economic Inequality*, 10(3), 421–443. https://doi.org/10.1007/s10888-011-9188-x

Conceição, P., & Ferreira, P. (2000). *The young person's guide to the Theil index: Suggesting intuitive interpretations and exploring analytical applications* (UTIP Working Paper No. 14). University of Texas Inequality Project. https://utip.gov.utexas.edu/papers/utip_14.pdf

Cowell, F. A. (2011). *Measuring inequality* (3rd ed.). Oxford University Press.

Executive Office of the President. (2021). *Executive Order 14008: Tackling the climate crisis at home and abroad.* Federal Register, 86 FR 7619 (January 27, 2021). Rescinded January 2025.

Federal Highway Administration. (2023, February 28). National electric vehicle infrastructure standards and requirements; final rule. *Federal Register*, 88 FR 12724. https://www.federalregister.gov/documents/2023/02/28/2023-03500/national-electric-vehicle-infrastructure-standards-and-requirements

Gini, C. (1912). *Variabilità e mutabilità: Contributo allo studio delle distribuzioni e delle relazioni statistiche.* C. Cuppini.

Hsu, C.-W., & Fingerman, K. R. (2021). Public electric vehicle charger access disparities across race and income in California. *Transport Policy*, 100, 59–67. https://doi.org/10.1016/j.tranpol.2020.10.003

Hwang, C. L., & Yoon, K. (1981). *Multiple attribute decision making: Methods and applications* (Lecture Notes in Economics and Mathematical Systems, Vol. 186). Springer-Verlag. https://doi.org/10.1007/978-3-642-48318-9

Saaty, T. L. (1980). *The analytic hierarchy process: Planning, priority setting, resource allocation.* McGraw-Hill.

13. Acknowledgments

The author thanks Dr. Majed Al-Ghandour, Faculty Advisor at Fayetteville State University's Broadwell College of Business and Economics, for methodological guidance during the development of this study.

The study relies on publicly available data from several federal and state programs. The author acknowledges the North Carolina Department of Transportation (Climate Change Initiatives: monthly BEV registration statistics; NEVI program announcements); the U.S. Department of Energy and the National Renewable Energy Laboratory (Alternative Fuels Data Center charging-station inventory and API access); the U.S. Census Bureau (American Community Survey 2022 five-year estimates, TIGER geographic products, and the Longitudinal Employer-Household Dynamics program for LEHD LODES origin-destination commuter data); the Council on Environmental Quality, Executive Office of the President (the original CEJST v2.0 methodology and Technical Support Document); and the Environmental Data and Governance Initiative (EDGI) together with the Public Environmental Data Partners coalition (PEDP) for archival preservation of CEJST v2.0 following the decommissioning of the federal screening tool on January 22, 2025.

Funding. This study received no external funding. It is self-directed research with no corporate sponsor.

Data and Code Availability. All analytical scripts, processed data outputs, raw datasets, figure-generation code, and validation suites used to produce this manuscript are publicly available at https://github.com/wolfieman/ev-pulse-nc under the Polyform Noncommercial 1.0.0 license. The repository's `data/README.md` provides a 19-entry provenance table documenting the source and pull date for each dataset; the `src/analysis/` directory contains all scripts that produced the figures and processed CSVs referenced in this manuscript. The analysis is reproducible from a clean clone using `uv` to install the dependency lockfile (`uv.lock`) and re-execute the published scripts against the committed raw data.

Conflicts of Interest. The author declares no conflicts of interest.

AI-Assisted Methodology Review. The study's data quality panel and manuscript advisory panel processes used AI-assisted role-play methodology to expand solo-researcher analytical capacity. This methodology is documented in detail in Appendix B, including disclosure of AI tools used, the verification chain that anchors each panel's outputs to specific repository commits, and the explicit distinction between internal methodology review and external peer review. AI tools are research and writing aids in this work and are not co-authors; sole authorship rests with Wolfgang Sanyer.

Appendix A — Data Quality Methodology Review (Summary)

This appendix is a one-page summary of an internal data quality methodology review conducted on the EV Pulse NC analytical pipeline. The full review is published as a companion document at `docs/research/data-quality-review-full.pdf` in the project repository, with a 20-slide visual summary at `docs/research/data-quality-review.pdf`; the full document includes a roughly 27,000-word section-by-section walkthrough of all six data quality dimensions, a 10-section roadmap traversal, a roughly 60-item disposition log, the 8-point Data Governance Readiness lineage chain, and a Claims-to-Evidence Matrix that ties every major numerical claim in this manuscript to a source file. The summary below establishes scope, panel composition, outcome, and the specific findings carried forward into the body of this manuscript.

A.1 Scope and Outcome

The data quality review was conducted in April 2026 against a fixed repository commit (`be2a27b`, April 10, 2026) and used the AI-assisted role-play methodology disclosed in Appendix B. The review covered approximately sixty distinct items organized across the six standard data quality dimensions (Accuracy, Completeness, Consistency, Timeliness, Validity, Uniqueness) and the ten sections of a comprehensive data quality roadmap (Business Objective, Data Requirements, Data Source Assessment, the Six Dimensions, Cleaning and Preparation, EDA for Quality Validation, Validate Assumptions, Governance and Documentation, Risk Assessment, and Iterate). Of the roughly sixty items, zero remained open at evaluation snapshot. Every item resolved to either a strength, a limitation with documented direction of bias, or a future-work pointer; no item was deferred or left ambiguous.

A.2 Panel Composition

The five-lane review brought five specialist perspectives to bear on the analytical pipeline.

Lane	Specialist scope	Primary contribution to this manuscript
1. Data Quality Engineering (DAMA-DMBOK)	Source provenance, dataset-level data quality posture, six-dimension audit	Source provenance audit; cleaning-log structure; six-dimension resolution (Section 4 framing, Section 5 governance content)
2. Spatial / GIS Data	Federal spatial standards (EPA EJScreen, HUD USPS Crosswalk), CRS discipline, area-weighted crosswalks	23-of-23 spatial validation suite; tract-to-ZCTA area-weighted crosswalk validation (Section 4.5)
3. Federal Data and Policy Compliance	CEJST archival pathway, federal-data continuity risk, Justice40 compliance posture	CEJST v2.0 archival pathway disclosure (Sections 5.5, 9.8); direction-of-bias disclosure framing (Section 9)
4. Econometrics and Forecast Validation	Out-of-sample validation design, structural-break testing, multicollinearity, residual diagnostics	Diagnostic suite consolidation (Section 4.9); Chow test framing (Section 6.1); VIF max 1.41 (Sections 4.9, 6.6)
5. Business and Data Analytics Strategy	Stakeholder mapping, archetype framing, weakness ordering, audit-readiness positioning	Three-archetype framing (Section 7.3); stakeholder-by-stakeholder value proposition (Section 7.2); weakness ordering (Section 9)

A.3 Selected Findings Reflected in This Manuscript

The review surfaced or confirmed a number of substantive findings that the manuscript body either adopts directly or references explicitly. A non-exhaustive list:

- **Multicollinearity check.** Maximum Variance Inflation Factor (VIF) across the three scoring pillars is 1.41, well below the conventional 5.0 threshold (cited in Manuscript §4.9 and §6.6).
- **CEJST archival pathway disclosure.** The federal CEJST tool was decommissioned on January 22, 2025; the EDGI / Public Environmental Data Partners archive is the operational source for v2.0 (Manuscript §5.5 and §9.8).

- **Federal spatial-standard alignment.** The 12-check three-tier spatial validation suite passed at 23-of-23 against EPA EJScreen and HUD USPS Crosswalk federal practice (Manuscript §4.5).
- **Direction-of-bias disclosure pattern.** The framing "a limitation with a direction is a disclosure; a limitation without one is a wound" anchors Manuscript §9 and originates in Lane 3 of this review.
- **Three-archetype framing.** The Union (utilization-driven) / Mecklenburg (equity-driven) / Orange (low-across-pillars) interpretation in Manuscript §7.3 originates in Lane 5 of this review.
- **Data Governance Readiness chain.** The 8-point lineage chain from raw bytes through final NEVI score, summarized in Manuscript §5.6, is detailed in the full review.

A.4 Full Report and Citation Form

The complete review is published as a self-contained document at `docs/research/data-quality-review-full.pdf` (with a 20-slide visual summary at `docs/research/data-quality-review.pdf`) in the project repository (https://github.com/wolfieman/ev-pulse-nc). The full report includes: the section-by-section walkthrough of the data quality roadmap, the six-dimension audit, the Data Cleaning Log, the spatial validation suite, the diagnostic suite, the governance asset inventory, and the risk-assessment / data-limitations statement that is condensed into Manuscript §9. Citation form: *Sanyer, W. (2026). EV Pulse NC — Data Quality Report. https://github.com/wolfieman/ev-pulse-nc/blob/main/docs/research/data-quality-review-full.pdf*

Appendix B — AI Methodology Disclosure (Summary)

This appendix is a one-page summary of the AI-methodology disclosure for this study. The full disclosure is published as a companion document at `docs/research/ai-methodology-disclosure.pdf` in the project repository; it covers the disclosure statement, the AI tool stack, the workflow-integration walkthrough, the author-AI boundary table, and the reproducibility implications. The summary below establishes the four minimum elements that the Committee on Publication Ethics (COPE), *Nature*, and *Science* converge on for AI-use disclosure: tools used, tasks they performed, the author's responsibility for the work, and a pointer to the full disclosure.

B.1 Tools Used

The study used six AI tools, each in a documented role.

- **Claude Code (Anthropic, CLI):** primary technical and analytical assistant. Code generation and refactoring for the Python analysis scripts, the role-play methodology that operationalized the data quality review (Appendix A) and the manuscript advisory outline review, source-read synthesis during manuscript preparation, and figure-generation code (fig-43 through fig-45).
- **Claude.ai (Anthropic, web app; "Claude Browser"):** writing-polishing layer and parallel-review second opinion on framing decisions.
- **ChatGPT (OpenAI, web app):** broad orchestration and advisory layer for high-level framing, decision support across competing methodological choices, and structured workflow capture across the broader research process.

- **Perplexity AI:** deep research and citation-finding for the literature review (Section 3), particularly the prior-work check on the Theil novelty claim that surfaced Choi, Xu, and Jiao (2025).
- **NotebookLM (Google):** document synthesis layer for grounded synthesis of research papers and primary-source materials identified through Perplexity.
- **Gamma:** used only for the separate competition-presentation deck (`ev-pulse-nc-prioritizing-investment.pptx`), not for the manuscript itself.

B.2 Tasks Performed With AI Assistance

AI tools accelerated four categories of task in this study. (1) **Code generation and refactoring** of the Python analytical pipeline, with outputs verified against source data by the Claims-to-Evidence Matrix in the data quality review (Appendix A and `docs/research/data-quality-review.pdf` §8). (2) **Structured role-play methodology review** in two forms: the five-lane data quality panel (Appendix A) and the four-panelist manuscript advisory outline review. Each was anchored to a fixed repository commit and to a pre-registered claims matrix. (3) **Literature-review filtering** via Perplexity and NotebookLM, with the primary sources read by the author where the literature-review claims depend on them. (4) **Prose drafting and polishing**, in which Claude Code drafted initial section prose against author-defined content commitments and Claude.ai polished the result; the author revised and accepted only what survived author review.

B.3 Author Responsibility Statement

Sole authorship of this study rests with Wolfgang Sanyer. AI tools used in this study are research and writing aids, not co-authors. The author defined the scope, the target audience, the methodological choices, the scoring weights, and the decision thresholds. The author verified every numerical claim against source data through the Claims-to-Evidence Matrix. The synthesis across analytical phases and across the methodology review process is the author's. So is every interpretive decision (county archetypes, two-tier architecture framing, scope decisions) and every methodological framing locked into the manuscript's claims. The author authored, signed, and accepts authorial responsibility for the manuscript and its conclusions. AI tools provided structured multi-perspective review, candidate drafts, candidate framings, and analytical outputs subject to author verification. None of that substitutes for the author's judgment or the verification chain.

This position aligns with the Committee on Publication Ethics (COPE) authorship guidance, with the *Nature* and *Science* policies on AI use in scholarly publications, and with the author's policy of disclosing AI use without crediting AI as an author.

Two limitations of the AI-assisted methodology are carried forward into the manuscript's Limitations section (§9). First, the data quality panel and the manuscript advisory panel processes constitute **internal methodology review**, not external peer review. Second, the verification chain is anchored to repository commits that the author can re-derive; no external audit has been performed.

B.4 Full Disclosure

The complete AI-methodology disclosure for this study is at `docs/research/ai-methodology-disclosure.pdf` (https://github.com/wolfieman/ev-pulse-nc/blob/main/docs/research/ai-methodology-disclosure.pdf). The full document covers: the disclosure statement and its alignment with COPE / *Nature* / *Science* conventions; the AI-tool-stack walkthrough with role definitions for each tool; the workflow-integration narrative across the five analytical phases plus the methodology review and manuscript drafting phases; the author-AI boundary table; and the reproducibility implications. The disclosure also documents the broader AI-workflow infrastructure that this study draws on, visible in the companion orchestrator repository at https://github.com/wolfieman/orchestrator. Citation form: *Sanyer, W. (2026). AI methodology disclosure — EV Pulse NC. https://github.com/wolfieman/ev-pulse-nc/blob/main/docs/research/ai-methodology-disclosure.pdf*